Selecting the Right Mode: Symmetry-Aware, Orbit-Consistent Tesseract Trees on Frozen Foundation Features for 6D Object Pose Estimation

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Abstract

Adapting a 6D pose estimator to a new object or application usually means fine-tuning a large network on new data. We study whether shared, frozen *foundation backbones can instead support accurate 6D pose of known objects across very different object types, with only small downstream heads trained per object. Frozen DINOv3 appearance features, optionally fused with frozen MoGe-2 surface-normal features, feed a compact head (2.4 M parameters per branch) that predicts rotation as a root-to-leaf path in a hierarchical tesseract partition of $SO(3)$, and a 0.33 M-parameter block that predicts translation from the same encoded tokens. A base head trains in a median of 5–16 minutes from cached features (distilled students take 40–84 minutes); the backbones (0.30 B and 0.33 B parameters) are never trained. Because single-view errors are dominated by selecting the wrong mode among symmetric or look-alike rotations, the heads use symmetry-aware multi-path supervision, orbit aggregation of hierarchical hypotheses over exactly mapped in-plane rotations, and optional distillation of that consistency into single-view students. With validation-selected configurations and ground-truth crops, the same architecture reaches a recall of 79.0% at 10° and 88.0% at 5 cm on real T-LESS images, 2.07° on SYMSOL-T, and 12.9° and 1.6 cm on YCB-V, where training on real videos alone fails; on T-LESS and YCB-V these rotation results use the ground-truth translation to convert allocentric predictions. Applied unchanged to spacecraft pose under a synthetic-to-real shift (SPEED+), it reaches $11.3^\circ/11.9^\circ$ on the real domains; with a synthetic-only detector instead of ground-truth crops the error doubles on one of them. Multi-path targets improve every symmetric benchmark and, unlike canonical labels, keep orbit aggregation effective; orbit aggregation helps on three of the four benchmarks but not on cluttered YCB-V; geometry fusion helps only on two. Absolute accuracy remains below fine-tuned state-of-the-art methods.*

1. Introduction

Many applications need the 6D pose of a *known* object from a single image: a robot picking a part from a bin, an inspection station checking the orientation of an assembly, a warehouse arm grasping a product, or a servicer spacecraft approaching a target. Each application brings its own objects, cameras, lighting and training data. Today, adapting a pose estimator to a new object or domain typically means fine-tuning a large network, which is costly in data, compute and engineering, and has to be repeated for every new application. Foundation backbones offer an alternative: their features already encode appearance and 3D geometry well enough that, in principle, they could be shared and kept frozen, with only a small downstream head trained for each new object. This paper asks:

> *Can shared frozen appearance and geometry backbones support accurate 6D pose estimation across different object types through compact, inexpensive-to-train downstream heads?*

We study this with a modular architecture. A frozen DINOv3 [22] backbone supplies appearance tokens and, optionally, a frozen MoGe-2 [24] backbone supplies surface-normal tokens. A standard interface pools these into fixed-size token grids, and per-object heads read them: a hierarchical rotation tree that predicts a path in a tesseract partition of $SO(3)$ [15], and a translation block that reads the same encoded tokens. The two branches are fused per tree level by a training-free product of experts (PoE) [9]. We call this design *plug-and-play* in a narrow sense: a new object reuses the backbones, the feature interface, the head architecture, the training recipe and the inference algorithm, and only its heads are trained, which takes minutes. It is *not* zero-shot: each object needs labelled training images in a fixed object frame, camera intrinsics, a crop (a ground-truth box or a trained detector), and its symmetry set if it is symmetric, and the heads of one object do not transfer to another. A few inference options (geometry fusion, orbit aggregation) are chosen per application on validation data.

The main difficulty for such small heads is *selecting* the right mode. A symmetric object has several rotations that produce the same image, all of them correct; a nearly symmetric object has look-alike rotations of which only one is correct; and under a domain shift the details that separate them become unreliable. Distributional pose methods [7, 13, 16] can represent such multi-modal posteriors, but a single prediction still has to select a mode. Hierarchical hypotheses make this selection explicit, and we address it at three points: *training* targets that cover all symmetry-equivalent rotations (Sec. 3.4); *orbit aggregation* at test time, which scores the tree's candidates for consistency across in-plane rotations of the crop through an exactly derived camera-roll transform (Sec. 3.6); and *orbit distillation* of that consistency into single-view students (Sec. 3.7).

We evaluate the same architecture on three general-object benchmarks that stress different kinds of difficulty: textureless industrial parts with continuous and discrete symmetries (T-LESS), exactly symmetric solids (SYMSOL-T) and cluttered household objects (YCB-V). We then apply it unchanged to a demanding downstream application, spacecraft pose estimation for autonomous rendezvous under a synthetic-to-real shift (SPEED+ [19]). The other applications named above motivate the design but are not evaluated here. All configurations are selected on validation data, all criteria were fixed before the runs, and results that rely on ground-truth crops or ground-truth translation are labelled as such.

Our contributions are as follows. (1) A reusable 6D pose architecture in which frozen, shared foundation backbones (0.30 B and 0.33 B parameters) feed per-object rotation and translation heads of 2.1–2.4 M and 0.33 M parameters, of which a base head trains in a median of 5–16 minutes from cached features (the distilled SPEED+ students take 40–84 minutes); we report the full adaptation and inference cost, including feature extraction and, where needed, detector training (Secs. 3.2 and 4.5). (2) A hierarchical rotation head whose size does not grow with the number of rotation hypotheses (131,072 leaves, 44 nodes scored per greedy decode), which is far more accurate than a flat classifier over SPACE-HOP's 131,076-anchor Hopf grid under the same recipe (3.87° vs. 7.53°) with 15× fewer parameters; a same-grid control that would separate the effect of hierarchy from that of the grid is still running (Sec. 5). (3) Symmetry-aware multi-path supervision for hierarchical rotation trees, which improves T-LESS at every recall threshold (orbit R@10 73.5% → 79.0%) and every SYMSOL-T shape (2.62° → 2.07°), and which, unlike canonical labels, keeps orbit aggregation effective (Secs. 4.3 and 5). (4) Orbit aggregation with an exact roll transform for hierarchical candidates and its distillation into single-view students: aggregation lowers validation error on T-LESS, SYMSOL-T and SPEED+ at about 4–5× inference cost (measured on SPEED+ crops), and on SPEED+ repairs 62–80% of severe flips, while distillation further improves the spacecraft system at no inference cost; aggregation hurts on cluttered YCB-V, so it is an option selected on validation rather than a universal gain (Secs. 4.4 and 5). (5) Cross-dataset findings that bound the shared design: translation from the same frozen tokens is far more accurate than the box alone on every benchmark (T-LESS 88.0% within 5 cm, close to LiePose at 5 and 10 cm; YCB-V 1.6 cm mean error), although it remains well behind LiePose on SYMSOL-T; geometry fusion helps only where appearance is ambiguous (SPEED+, T-LESS) and costs about 4×; household objects in clutter require synthetic training data; and with a synthetic-only detector the spacecraft error doubles on one real domain, so detection, not the heads, limits its deployment there (Sec. 4). Symmetry-aware losses, test-time augmentation and self-distillation are established ideas; our contribution is their formulation for hierarchical rotation heads on frozen features and the measured evidence above. Fine-tuned state-of-the-art methods remain more accurate on every benchmark we use.

2. Related Work

Object pose estimation. Learning-based 6D pose estimators regress poses directly or predict 2D–3D correspondences followed by PnP [25]. The BOP benchmark [11] standardises evaluation on datasets such as T-LESS [10] and YCB-V [25], where textureless and symmetric objects remain the hardest cases. Most of these methods train or fine-tune a network per dataset or per object set. We instead keep the foundation backbones frozen and shared, and ask how far small, selection-aware heads can take them.

Spacecraft pose as an application. Synthetic-to-real transfer is studied in its purest form on SPEED+ [19], where models train on renders of a spacecraft and are tested on hardware-in-the-loop captures. There, keypoint-and-PnP pipelines with fine-tuned backbones lead [17, 18, 20, 23]; the most accurate recent result fine-tunes DINOv3 with LoRA and reports that a head on *frozen* DINOv3 features fails with frequent flips [3]. Direct regressors are faster but less accurate [1, 5, 26]. We use this benchmark as a demanding downstream application of the general design, not as its target.

Discrete and distributional rotation prediction. Uniform $SO(3)$ grids can be built from the Hopf fibration [27] or by recursive tesseract subdivision [15]; we adapt the latter to cell-centre representatives with unique paths. Grid- and implicit-density methods model multi-modal rotation posteriors, through MLP scores over grids [12, 16] or equivariant spherical features [14]; parametric alternatives include Bingham mixtures [4]. SpyroPose [7] evaluates a hierarchi-

cal $SE(3)$ pyramid coarse to fine, and LiePose [13] applies score-based diffusion on $SE(3)$ and handles symmetric objects well. Continuous rotation regression has known representation issues [28]. Riemannian [2] and discrete [6] flow matching provide generative alternatives; we tested both as refiners and report them as negative results.

Consistency, test-time augmentation and distillation. Averaging predictions over transformed inputs is a standard route to test-time robustness, and four-rotation test-time augmentation is also used for spacecraft keypoints [3]. Data distillation [21] turns multi-transform ensembles into training targets, a form of knowledge distillation [8]. Equivariant architectures build rotation consistency into the network [14]. Our orbit voting is a test-time augmentation over in-plane rotations, and orbit distillation is a data-distillation variant. What is specific here is the exact camera-roll transform for crops of off-axis objects, scoring with hierarchical tree likelihoods rather than averaging point estimates, and the target: a diagnosed mode-selection failure.

3. Method

Figure 1 gives an overview of training and inference.

3.1. Problem setup

Given a square crop of a known object around its bounding box (15% margin per side), we predict the rotation $q \in SO(3)$ (unit quaternion, $q \equiv -q$) of the object frame relative to the camera and the translation $t \in \mathbb{R}^3$. For objects that appear away from the optical axis (T-LESS, YCB-V), the tree predicts the *allocentric* rotation $R_{\text{ray}}(t)^\top R$, which removes the viewing-ray component so that the crop's appearance depends only on the predicted rotation; it is converted back exactly. Rotation error is $E_R = 2 \arccos |\langle \hat{q}, q \rangle|$, minimised over the object's symmetry set $\mathcal{S}$ when one exists ($\min_{S \in \mathcal{S}} E_R(\hat{q}, q\,S)$), and translation error is $E_T = \|\hat{t} - t\|_2$.

Use of ground-truth translation at test time. The allocentric conversion needs the viewing ray of the object, *i.e.* its translation. In the results reported here for T-LESS and YCB-V, the conversion and the mapping of orbit candidates between rotated views use the *ground-truth* translation; the reported allocentric error equals the egocentric error only under that ray. An evaluation that substitutes the predicted translation (Sec. 3.8) for both uses is pre-registered and not yet complete. SPEED+ and SYMSOL-T use egocentric targets and roll about the crop-centre ray, so their rotation results use no translation.

3.2. What is shared, supplied and trained

The architecture separates a shared part from a per-object part. *Shared and frozen* for every application: the DINOv3 ViT-L/16 backbone (303 M parameters) and, optionally, MoGe-2 ViT-L (331 M), the feature interface (pooled token grids of fixed shape: 64×1024 for DINOv3 and 256×4 for MoGe-2, standardised with training statistics), the head architecture, the training recipe and the inference algorithm. *Supplied per object*: the object identity at test time (to select its heads), training images with poses in a fixed object frame, camera intrinsics, a square crop (here the ground-truth box; for deployment, a detector, which must itself be trained, Sec. 4.4), and the object's symmetry set if it has one. Training images may be synthetic; for cluttered household objects, renders were necessary (Sec. 4.3). *Trained per object*: one rotation tree per backbone branch (2.39 M parameters for DINOv3, 2.13 M for MoGe-2) and a translation block (0.33 M). Features are extracted once per image and view and cached, so a head trains in minutes (Sec. 4.5). *Chosen per application on validation data*: whether to fuse the MoGe-2 branch (PoE) and whether to use orbit aggregation. Nothing is chosen on test data. The result is a shared architecture with per-object heads, not a single universal predictor: heads trained for one object do not transfer to another (Appendix H).

3.3. Tesseract tree on $SO(3)$

Following the recursive tesseract subdivision [15], a quaternion is assigned to a *chart* $c = \arg\max_i |q_i|$, sign-normalised so that $q_c > 0$ (this identifies q and $-q$), and projected to the cube $x = q/q_c$. Its three free coordinates in $[-1, 1]$ are halved recursively, so each level selects one of 8 children. We use *cell centres* (lifted back to the unit sphere) as representatives. Unlike corner representatives, every rotation then has a unique root-to-leaf path. With 4 charts and depth 5 there are $4 \cdot 8^5 = 131{,}072$ cells. The quantisation error of the leaf centre is 2.51° on average (3.97° p95, 6.20° worst case), and it halves per level. For objects without symmetry this floor bounds the fine accuracy of our rotation results; for symmetric objects the nearest leaf to *any* equivalent rotation is closer.

Tree Transformer. For each backbone, a small decoder predicts the path autoregressively. The frozen tokens are layer-normalised, linearly projected to $d = 256$, given 2-D positional encodings and passed through one encoder layer. A causal decoder with 2 layers (4 heads, FFN 512) then reads a sequence that starts with a root token followed by one token per decided level. Each such token embeds the level and a 6-frequency Fourier encoding of the current cell centre, and cross-attends to the encoded image tokens. The root position predicts the chart (4 logits) and position ℓ predicts the child (8 logits). The model has 2.4 M parameters. We train with geodesic soft targets: at each level, the target over the candidate cells is $\propto \exp(-(\delta/\tau_\ell)^2)$, where δ is the angle between the ground truth and each cell centre. The

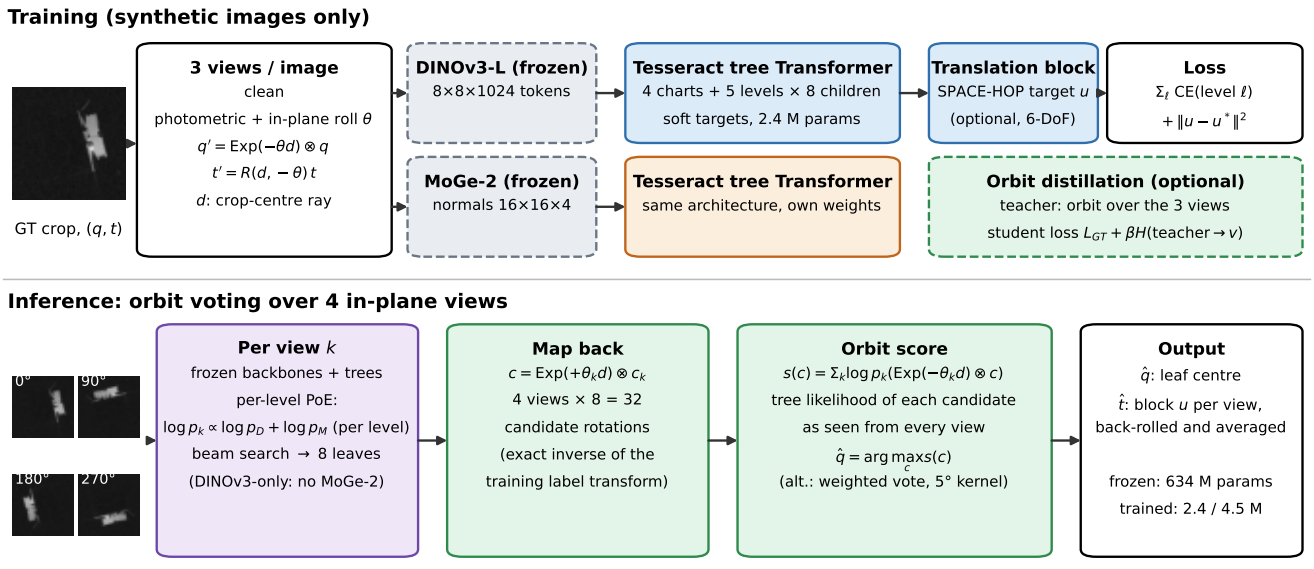


Figure 1. **Overview.** *Top:* training. Each synthetic crop is used as a clean view and as two augmented copies (photometric augmentation plus an in-plane roll θ with exactly transformed labels). Frozen DINOv3 and MoGe-2 features feed one tesseract tree Transformer each; an optional translation block reads the same tokens, and optional orbit distillation replaces the ground-truth target by an orbit teacher. *Bottom:* orbit voting at test time. Candidates from four rotated views are mapped back to a common frame and scored by their per-level PoE tree likelihood in every view.

widths τ_ℓ ($39.3°$ for the root and level 1, $20.0°$, $10.0°$, $5.0°$, $2.5°$) halve with the cell size. Training uses teacher forcing on the ground-truth path and a cross-entropy at every level; inference is greedy or beam search, and the prediction is the leaf centre.

Frozen features. DINOv3 ViT-L/16 [22] last-layer patch tokens are average-pooled to an 8×8 grid (64×1024). For geometry, MoGe-2 [24] surface normals are masked by a foreground mask on which both backbones agree (the intersection of a DINOv3 foreground estimate and the MoGe-2 mask). They are then pooled to 16×16 cells, each holding three masked-normal channels and the mask fraction (256×4). This consensus mask removes most of MoGe-2's background leakage on real images (Appendix B). Features are extracted once; only the trees are trained.

Per-level product of experts. The two trees are fused without training. At every decoding step, the log-softmax distributions of both branches are added and renormalised: $\log p^{(\ell)} \propto \log p^{(\ell)}_{\text{DINO}} + \log p^{(\ell)}_{\text{MoGe}}$. Beam search then runs on the fused tree. We report this *PoE* variant and a faster *DINOv3-only* variant.

3.4. Symmetry-aware multi-path targets

For a symmetric object, the annotated rotation q is one arbitrary member of the equivalence set $\{q\,s : s \in \mathcal{S}\}$, where $\mathcal{S}$ holds the object-frame symmetries (continuous symmetries discretised). Training on q alone (*single label*) gives contradictory targets for identical images. The usual fix maps every label to a *canonical* member, *e.g.* the one closest to identity. This removes the contradiction but makes the target a discontinuous function of the image: two nearly identical views can receive canonical labels far apart, and the in-plane relation of Eq. (2) no longer holds between views. Instead, we train the tree on all equivalent rotations. The loss is the soft-target cross-entropy of Sec. 3.3, averaged over the members of the set, each with teacher forcing along its own path:

$$\mathcal{L}_{\text{sym}} = \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} \mathcal{L}_{\text{GT}}(q\,s), \tag{1}$$

where $\mathcal{S}' = \mathcal{S}$ if $|\mathcal{S}| \le 8$ and otherwise 8 members are drawn per sample and step (an unbiased estimate). The tree learns to split its probability between the equivalent paths, which is the level-marginal target of the whole set. For objects without symmetry Eq. (1) reduces exactly to the standard loss. Right-multiplication by s commutes with the roll of Eq. (2), so augmented views and orbit voting are unchanged; we verified both identities numerically ($< 10^{-9}$).

3.5. Augmentation with exact in-plane labels

Each training image is used as a clean crop and as two pre-extracted augmented copies; one of the three is drawn at every step. The augmentations are photometric (brightness, contrast, gamma, blur, glare blobs, noise, random erasing, JPEG) plus an in-plane rotation of the crop by $\theta \sim \mathcal{U}(-180°, 180°)$ about its centre. Rotating the crop about its centre is equivalent to rotating the camera about the unit ray d through the crop centre, computed with the lens distortion removed. Points therefore transform as $p' = R(d, -\theta)\, p$, and the labels as

$$q' = \mathrm{Exp}(-\theta\, d) \otimes q, \qquad t' = R(d, -\theta)\, t. \tag{2}$$

We derived this convention from the camera geometry rather than fitting it to a predictor. Projecting model keypoints with q' reproduces the warped image to $0.2°$ (p95 label error measured by PnP). For datasets whose objects lie far from the optical axis (T-LESS, YCB-V), an affine crop rotation is not exactly a camera roll (label error up to $6.7°$). There we warp with the roll homography $KR(d, -\theta)K^{-1}$ instead, which makes Eq. (2) exact (verified to $0.000°$).

3.6. Orbit voting

At test time we form $K{=}4$ views of the crop rotated by $\theta_k \in \{0°, 90°, 180°, 270°\}$. In view k the fused tree's beam search returns 8 leaves $c_{k,1..8}$, which are mapped back to the original frame by the inverse of Eq. (2), $c = \mathrm{Exp}(+\theta_k d) \otimes c_k$, giving 32 candidates. Each candidate c is then scored by the tree likelihood of its own path when mapped into every view:

$$s(c) = \sum_{k=1}^{K} \log p_k\big(\mathrm{path}(\mathrm{Exp}(-\theta_k d) \otimes c)\big), \tag{3}$$

and predict $\hat{q} = \arg\max_c s(c)$. Here $\log p_k$ sums the fused root and child log-probabilities along the path, evaluated with teacher forcing. A $180°$ look-alike that wins in one view must also be likely in the other three to be selected, whereas the true pose is consistent by construction. We also evaluate a *weighted vote*, in which each candidate collects Gaussian-kernel support ($5°$) from the other candidates, weighted by their normalised beam probabilities within their view. Orbit voting has no parameters and multiplies inference cost by about $4\times$ (Tab. 7).

3.7. Orbit distillation

Orbit voting is more accurate than a single view, so we use it as a teacher. For every training image we run the frozen orbit system over its three available views (clean and two augmented, with known rolls). This gives 24 candidates in the clean frame with weights $w_j = \mathrm{softmax}_j\, s(c_j)$. A new tree per branch is then trained from scratch with the same recipe and the loss

$$\mathcal{L} = \mathcal{L}_{\mathrm{GT}}(v) - \frac{\beta}{L{+}1} \sum_j w_j \sum_{\ell=0}^{L} \log p^{(\ell)}\big(\mathrm{pre}_\ell(\mathrm{Exp}(-\theta_v d) \otimes c_j)\big), \tag{4}$$

with $\beta = 1$ and pre_ℓ the length-ℓ path prefix. Here v is the view drawn for the image, and the teacher candidates are transported into its frame. The student is a single-view model with unchanged inference cost; at test time it can be used alone or with orbit voting. This is a data-distillation scheme [21] applied to hierarchical rotation targets.

3.8. Translation block (6-DoF)

For translation we keep the SPACE-HOP parameterisation [5] unchanged. With box centre (b_x, b_y) and size (b_w, b_h), image size (W, H), intrinsics (f_x, f_y, c_x, c_y) and $\alpha = 1.6$, the target is $u = (U_x, U_y, U_z)$ with

$$U_x = \tfrac{1}{b_w}\big(f_x \tfrac{X}{Z} + c_x - b_x\big), \quad U_y = \tfrac{1}{b_h}\big(f_y \tfrac{Y}{Z} + c_y - b_y\big), \tag{5}$$

$$U_z = \tfrac{1}{2}\big(\tfrac{\alpha b_w}{W} + \tfrac{b_h}{H}\big) Z, \tag{6}$$

trained with $\|u - u^*\|^2$ and inverted in closed form. A translation block (one learned query cross-attending to the tree's encoded tokens, then an MLP) predicts u. It is trained either jointly with the tree ($\mathcal{L}_{\mathrm{tree}} + \mathcal{L}_{\mathrm{trans}}$) or on a frozen tree. On SPEED+ we evaluate both; the 6D system pairs the rotation of the distilled students with the jointly trained DINOv3 block. On T-LESS, SYMSOL-T and YCB-V the block is trained on the frozen, validation-selected rotation tree, so adding translation never changes the rotation results; each dataset uses its own image size in U_z, and u is standardised per component with training statistics. Rolled views use t' from Eq. (2) with the unchanged box. With orbit voting, each view's $\hat{t}$ is rolled back and the four are averaged.

4. Experiments

4.1. Benchmarks and protocol

Benchmarks. We use three general-object benchmarks that stress different kinds of ambiguity, and one downstream application. Every benchmark uses the same frozen backbones, the same head architecture, training recipe and inference algorithm; only the heads are trained. **T-LESS** [10]: textureless industrial parts with continuous, discrete or no symmetries. We use 7 objects spanning these classes (1, 4, 5, 11, 18, 21, 27), train on PBR renders and real training images, select on held-out PBR scenes, and test on all real

BOP19 targets [11] of these objects. **SYMSOL-T**: five symmetric solids (tetrahedron, cube, icosahedron, cone, cylinder) with random translations, rendered with LiePose's generator and seeds [13] (so the images follow its protocol but are not guaranteed identical to its published sets). **YCB-V** [25]: household objects in cluttered real videos. We use 6 objects without declared symmetries (cracker box, mustard bottle, banana, mug, power drill, scissors), validate on held-out real sequences and test on the real keyframes. **Application: SPEED+** [19]: spacecraft pose estimation for autonomous rendezvous, with a nearly symmetric target and a hard synthetic-to-real shift: 47,966 synthetic training and 11,994 synthetic validation images, tested on two real domains, *lightbox* (6,740) and *sunlamp* (2,791). Training uses synthetic images only.

Protocol. Experiments followed pre-registered protocols: question, arms, controls and pass criterion were fixed before each run, and protocol changes were recorded before the affected runs. Post-hoc analyses are labelled as such. We train 3 seeds and report mean $\pm$ seed standard deviation; a criterion passes if the difference exceeds the larger seed standard deviation. Checkpoints, hyper-parameters and variants are selected on validation data only; real test sets are never used for selection. One tree per object and branch is trained with the same recipe on every dataset. Two inference options are chosen per application on its validation data, never on test data: the variant (PoE or DINOv3-only) and whether to use orbit voting. Errors are averaged per object and then over objects. Unless stated otherwise, crops come from ground-truth boxes (Sec. 6). Failed criteria and negative results are reported (Appendix C).

Implementation. Every head is trained for 100 epochs with batch 256, AdamW (learning rate 10^{-3}, weight decay 0.05, one-cycle schedule) and bf16 on cached features; training and feature-extraction times are in Sec. 4.5. Timings use one shared NVIDIA RTX 6000 Ada and are upper bounds.

4.2. Results across applications

Table 1 summarises what the shared architecture achieves with the configuration each application selects on its validation data. Three things are worth noting. First, the selected configuration differs between applications: the geometry branch (PoE) is selected on T-LESS and SPEED+, the appearance branch alone on SYMSOL-T and YCB-V, and orbit aggregation on all applications except YCB-V. Second, translation predicted from the same frozen tokens is far better than the box alone on every benchmark (1.6–2.6 cm on the household and industrial objects), but it remains about 4–5$\times$ behind LiePose on SYMSOL-T and is limited by the box on detected SPEED+ crops. Third, two evidence boundaries matter: on T-LESS and YCB-V the rotation results use the ground-truth translation for the allocentric conversion (and, on T-LESS, for orbit mapping), and on SPEED+ the error doubles on one real domain when ground-truth crops are replaced by a synthetic-only detector. The following subsections compare each benchmark with published results.

4.3. General objects

This section compares the shared design with published results on the three general-object benchmarks.

T-LESS (Tab. 2). Orbit voting raises recall at all thresholds for the single-label trees (R@10 68.1 $\rightarrow$ 73.5%), and canonical labels help the single view but break orbit voting (Sec. 5). Multi-path targets (A2) are best at every threshold, with or without orbit voting: with orbit voting, R@2/5/10 rise to $5.7/48.1/79.0\%$ and the mean error falls to $15.2°$, and on validation the orbit error drops from $9.25°$ (A0) to $8.28°$, which passes the pre-registered criterion. Unlike canonical labels, multi-path targets keep orbit voting effective. LiePose is clearly more accurate, especially at fine thresholds (R@2 34.9% vs. 5.7%). Three factors are confounded in this gap: LiePose fine-tunes its backbone on all 30 objects, it masks the crop with the ground-truth visible mask, and our depth-5 leaves quantise rotations to $2.5°$ on average, which caps R@2.

SYMSOL-T (Tab. 3). Multi-path targets lower the error on every shape, from $2.62°$ to $2.07°$ on average. For symmetric shapes this is below the $2.31°$ error of the ground-truth rotation's own leaf, because the tree may pick the leaf closest to *any* equivalent rotation. LiePose remains about $4\times$ more accurate; closing the gap needs a continuous refinement on top of the selected leaf.

YCB-V (Tab. 4). Trained on the real training videos alone, the trees fit their training frames but memorise video context and fail on new sequences (test 69–81°; Appendix G). Adding the standard synthetic renders, pasted onto real training-sequence backgrounds, turns this into a working model. The configuration selected on validation (DINOv3, single view, $9.0°$ validation) reaches $12.9°$ on the real test keyframes, with 85% of predictions within $10°$; the PoE single-view model is better on test ($10.9°$, 87%) but is not the validation choice. Five of the six objects reach 4–8°, and the thin, near-planar scissors remain the main failure (Tab. 22). Unlike on the other benchmarks, orbit voting does *not* help on YCB-V: it raises the validation error from $9.0°$ to $11.5°$ (DINOv3) and from $10.5°$ to $12.7°$ (PoE). We have not yet diagnosed why; one hypothesis is that, with cluttered real backgrounds and a narrow range of camera roll in the training videos, the rotated views are scored less reliably than the original view (Appendix J.3). Published YCB-V results

Application	Train data	Crop	Selected (val)	E_R (°)	$\leq 10°$ / R@10	E_T	6D
T-LESS (7 obj.)	PBR + real	GT box	PoE, orbit	15.2	79.0%	2.6 cm	‡
SYMSOL-T (5 shapes)	synthetic	full frame	DINOv3, orbit	2.07	100%	0.061 u	–
YCB-V (6 obj.)	real + syn.	GT box	DINOv3, 1 view	12.9	85%	1.6 cm	‡
SPEED+ (LB/SL)	synthetic	GT box	dist. PoE, vote	11.3/11.9	86/79%	0.18/0.18 m	0.229/0.239§
		detector		25.0/16.9	–	0.66/0.27 m	0.541/0.341§

Table 1. **One architecture, four applications**: real test sets, validation-selected configuration per application, 3 seeds. E_R: mean symmetry-aware rotation error (T-LESS, SYMSOL-T; YCB-V and SPEED+ objects have no declared symmetry, so it equals the raw error). $\leq 10°$ / R@10: share of test predictions within $10°$. E_T: mean translation error of the validation-selected DINOv3 translation block on the frozen rotation tree (paired with PoE rotation on T-LESS; SYMSOL-T in scene units; SPEED+: jointly trained DINOv3 block). 6D: joint success. ‡ T-LESS and YCB-V rotation results convert the allocentric prediction and map orbit candidates with the ground-truth translation; the evaluation with predicted translation and joint 6D success is pending. § official SPEED+ score (lower is better). Ground-truth crops except the detector row. LB/SL: lightbox/sunlamp.

Method	Inference	R@2	R@5	R@10	mean (°)
LiePose [13]	1 sample	34.9	78.2	88.9	–
Ours, A0	1 view	2.3	27.0	68.1	17.9
	orbit (4)	4.0	36.0	73.5	16.6
Ours, A1	1 view	3.6	33.8	73.4	16.7
	orbit (4)	2.7	26.7	65.9	18.2
Ours, A2	1 view	4.5	40.6	76.5	15.7
	orbit (4)	5.7	48.1	79.0	15.2
Ours, A2 + distill.	1 view				
	orbit (4)				

Table 2. **T-LESS**, real BOP19 test targets of our 7 objects. Symmetry-aware rotation recall (%) at 2/5/10° and mean error, PoE variant, 3 seeds. LiePose numbers are its per-object results (Tab. 8 of [13]) averaged over the same 7 objects; it fine-tunes its backbone jointly on all 30 objects and multiplies the crop by the ground-truth visible mask. Ours: frozen backbones, ground-truth box, no mask; continuous symmetries (objects 1 and 4) are discretised in $10°$ steps for the error, which can add up to $5°$ for those objects. A0/A1/A2: single, canonical and multi-path targets (Sec. 3.4). Blank cells: runs not yet completed.

Method	tet	cube	icosa	cone	cyl	avg.
LiePose [13]	0.59	0.58	0.64	0.54	0.41	0.55
Ours, A0 + orbit	3.06	2.90	3.71	1.79	1.65	2.62
Ours, A2 + orbit	2.46	2.30	2.52	1.57	1.48	2.07
Ours, A2 + distill. + orbit						

Table 3. **SYMSOL-T** test, symmetry-aware mean rotation error (°) per shape. Ours: DINOv3 variant (selected on validation), orbit voting, 3 seeds. LiePose: SE(3) model, Tab. 3 of [13]. Blank cells: runs not yet completed.

use BOP or ADD(-S) metrics on all 21 objects with detected boxes, so we do not place our numbers in those tables.

Translation (Tab. 5). The translation block is the same on every dataset (Sec. 3.8) and is trained on the frozen tree selected for rotation, using each dataset's own image size. Its

Training data	Feat.	Inference	val	test	test acc@10
real videos	DINOv3	1 view	64.3 ±11.6	71.2 ±7.0	8%
	DINOv3	orbit (4)	63.0 ±2.4	69.4 ±5.2	9%
	PoE	1 view	73.5 ±6.8	81.1 ±7.3	16%
	PoE	orbit (4)	74.1 ±7.4	79.9 ±3.2	18%
+ synthetic (ours)	DINOv3	1 view	9.0 ±1.1	12.9 ±6.9	85%
	DINOv3	orbit (4)	11.5 ±1.8	13.4 ±4.7	83%
	PoE	1 view	10.5 ±0.9	10.9 ±4.1	87%
	PoE	orbit (4)	12.7 ±1.0	12.2 ±5.5	84%

Table 4. **YCB-V**, 6 objects, rotation error (°) on held-out real sequences (val) and real keyframes (test), and the share of test predictions within $10°$. Adding synthetic renders composited on real training backgrounds turns a failure into a working model.

Dataset	Method	T@2	T@5	T@10	mean E_T
T-LESS	LiePose [13]	68.0	88.4	95.2	–
	box only	26.6	58.5	79.8	6.28
	ours, 1 view	52.7	86.2	96.2	2.79
	ours, orbit mean	57.3	88.0	96.7	2.57
YCB-V	box only	16.0	44.7	76.7	8.14
	ours, 1 view	73.0	96.0	99.9	1.57
	ours, orbit mean	68.8	92.9	99.6	1.83
SYMSOL-T	box only	–	–	–	0.958
	ours, 1 view	–	–	–	0.083
	ours, orbit mean	–	–	–	0.061

Table 5. **Translation on general objects**, real test sets, ground-truth boxes. Translation recall (%) within 2/5/10 cm and mean error (cm; SYMSOL-T in scene units). Box only: the translation implied by the ground-truth box and the training-mean depth scale; ours: the DINOv3 translation block on the frozen selected tree (Sec. 3.8), from the original view or averaged over the 4 orbit views. 3 seeds. LiePose: average over the same 7 T-LESS objects (Tab. 8 of [13]). SYMSOL-T is in scene units, so only the mean distance is reported. Blank cells: runs not yet completed.

evaluation on T-LESS, SYMSOL-T and YCB-V was pre-registered together with these rotation results. On YCB-

V the block passes its criterion clearly: on validation, the orbit-averaged error is 3.4 cm against 10.7 cm for the box alone. On the real test keyframes, the single-view block places 73% of the objects within 2 cm and 96% within 5 cm (mean 1.6 cm, against 8.1 cm for the box alone). As for rotation, orbit averaging does not help on YCB-V (1.8 cm), and validation selects the single view. On T-LESS the block also passes (validation 1.85 cm with orbit averaging against 4.95 cm for the box alone), and here orbit averaging helps. On the real BOP19 test targets it places 57.3/88.0/96.7% of the objects within 2/5/10 cm (mean 2.6 cm, against 6.3 cm for the box alone). This is close to LiePose at 5 and 10 cm (88.4/95.2%) and below it at 2 cm (68.0%), although LiePose fine-tunes its backbone and masks the crop with the ground-truth visible mask. On SYMSOL-T, where the crop is the full frame and the box carries no position information, the block reduces the mean error from 0.958 to 0.061 scene units with orbit averaging; LiePose reports 0.011–0.016. Orbit averaging of translation helps on T-LESS and SYMSOL-T and hurts on YCB-V, the same pattern as for rotation. On the spacecraft application, translation is evaluated with both joint and frozen training (Sec. 4.4).

4.4. Application: spacecraft pose under synthetic-to-real shift

We now apply the same design, unchanged, to spacecraft pose estimation for autonomous rendezvous. Only renders of the spacecraft are available for training, and the test images come from two hardware-in-the-loop facilities. The target is nearly symmetric, so this application stresses exactly the look-alike failure mode that orbit voting targets.

Comparison with published results (Tab. 6). The SPEED+ system combines the rotation of the distilled PoE students with orbit aggregation and the jointly trained DINOv3 translation block. Synthetic validation selects the weighted vote as the orbit scoring rule (3.27° vs. 3.69° for the likelihood product fixed in our pre-registered recipe), so we report the vote as the primary result and the product as the pre-registered alternative. With ground-truth crops, the selected system reaches 11.35°/11.91° on lightbox/sunlamp (11.21°/11.09° with the product), close to recent direct methods (PAID-ViT 12.3°/11.8°) and well ahead of a head on frozen DINOv3 features in the same crop setting (15.5°/24.5° [3]). Fine-tuned keypoint pipelines are far more accurate (1.2°–3.4°). The official score is 0.229/0.239 (0.227/0.224 with the product; 0.241/0.238 for the jointly trained 6-DoF PoE model), similar to SPNv2 without adaptation on sunlamp and to PAID-ViT, and dominated by the rotation term. With boxes from a detector trained on synthetic images only (YOLO11s, trained in 6.3 h; a missed detection falls back to the full image and is still scored), the selected system reaches 24.96°/16.90° and a score of 0.541/0.341 (23.88°/16.42° and 0.523/0.333 with the product). The detector is reliable on synthetic images (mean IoU 0.99) and on sunlamp (0.88, no misses) but not on lightbox (mean IoU 0.76, 16% of boxes below IoU 0.5, 54 misses), so the lightbox gap is mainly a detection problem. Translation suffers most, because our translation target is defined relative to the box (0.66/0.27 m against 0.19/0.18 m with ground-truth boxes). This matched, synthetic-only setting is far behind SPNv2 without adaptation (0.142/0.230), which predicts from the full image.

Diagnosis: real-domain errors are flips. Before changing the model, we audited the large real-domain errors of the clean (non-augmented) SPEED+ trees. Encoder and decoder round trips are exact, and the metric, rotation convention and data pipeline were ruled out. Chart seams were ruled out as well: flat Hopf-grid heads on the same features flip as often or more. Of the errors above 90°, 36–41% lie within 15° of 180°, against 20% for uniformly random wrong rotations. Their body-frame rotation axes cluster around a few axes of the near-symmetric spacecraft. Allowing a 180° rotation about one body axis lowers the lightbox mean of the DINOv3 tree from 38.9° to 23.5°. The correct mode is usually available: among the tree's beam candidates and their 180° flips, an oracle achieves 9.8° on lightbox, against 26.9° for the fused greedy prediction. Neither a continuous flow posterior, flow-based re-noise scoring, synthetic templates nor a discrete path flow selects it on real images (Appendix C). The problem is *mode selection* under domain shift.

Orbit voting and distillation. Table 7 summarises the progression. Augmentation alone (photometric plus roll, single view) cuts the real-domain error of the PoE system from 26.9°/36.7° to 17.3°/20.2°. Orbit voting then brings it to 12.3°/12.3° and the synthetic error to 3.87°. It repairs 62% and 80% of the severe flips on lightbox and sunlamp while breaking only 0.4% of good predictions (Tab. 18). The median errors on real images are 5.3° and 6.3°, so the remaining means are driven by a tail of failures. Orbit distillation improves every setting. The single-view students recover 13–43% of the orbit gain at single-view cost; for example, the PoE greedy error on lightbox drops from 17.3° to 15.9°. Students with orbit aggregation give the best system: with the pre-registered likelihood product, 3.69° synthetic and 11.21°/11.09° on lightbox/sunlamp, about twice the seed std better than the undistilled orbit system, with fewer severe flips; with the validation-selected weighted vote, 3.27°/11.35°/11.91°. Synthetic validation ranks the distilled and undistilled systems in the same order as the real domains, but it does not rank the two scoring rules as the real domains do (Sec. 5). MoGe-2 helps on real images: PoE with orbit voting reaches 12.3° on lightbox, against 17.4°

Method	Setting	E_R (°) LB	E_R (°) SL	score LB	score SL
TangoUnchained* [20]	detector + keypoints/PnP, real images	3.19	4.30	0.073	0.090
lava1302* [20]	detector + keypoints/PnP, self-training	6.66	2.73	0.165	0.059
SPNv2 [17]	full image, no adaptation	6.48	11.07	0.142	0.230
SPNv2 + ODR* [17]	+ test-time adaptation	5.58	9.79	0.122	0.197
EagerNet [23]	detector + dense correspondences	1.75	2.66	0.039	0.059
SPNv3-B† [18]	GT crop, keypoints/PnP, fine-tuned	2.03	3.40	0.047	0.074
Church & Nikolian† [3]	GT crop, DINOv3-H+ LoRA, 12 passes	1.17	1.56	0.027	0.038
frozen DINOv3-L + head [3]	GT crop, frozen, 1 pass	15.47	24.49		
FastPose-ViT [1]	detector, direct regression	12.26	18.28		
PAID-ViT [26]	crop, direct regression	12.29	11.76	0.237	0.230
Ours, val-selected (vote)	GT crop, frozen, 4 views	11.35	11.91	0.229	0.239
	synthetic-only detector, frozen, 4 views	24.96	16.90	0.541	0.341
Ours, pre-registered (product)	GT crop, frozen, 4 views	11.21	11.09	0.227	0.224
	synthetic-only detector, frozen, 4 views	23.88	16.42	0.523	0.333
Ours, joint 6-DoF PoE (product)	GT crop, frozen, 4 views	12.11	11.73	0.241	0.238

Table 6. **SPEED+ real domains** (lightbox LB, sunlamp SL): mean rotation error and the official SPEED+ score (rotation error in radians plus normalised translation error, averaged over images; lower is better). Published values as reported by each source. * uses real or unlabelled test-domain images during training or adaptation. † ensemble of 3 models. Settings differ (detector or ground-truth crop, fine-tuned or frozen backbone), so only rows with the same setting are matched comparisons. Blank cells: runs not yet completed.

Training	Feat.	Inference	Synth. val	Lightbox	Sunlamp	>150° (%) LB	>150° (%) SL	ms
			mean E_R (°), 3 seeds					
Clean tree (no augmentation)	PoE	1 view	5.54 ±0.07	26.91 ±0.78	36.66 ±0.93	5.9	7.4	64
Augmented tree (base recipe)	DINOv3	1 view	4.84 ±0.05	22.55 ±0.73	22.00 ±0.70	6.0	3.2	15
Augmented tree (base recipe)	DINOv3	orbit (4)	4.21 ±0.04	17.40 ±0.68	15.27 ±0.64	4.8	2.0	70
+ orbit distillation	DINOv3	1 view	4.57 ±0.05	21.61 ±0.82	20.55 ±0.94	5.8	3.1	15
+ orbit distillation	DINOv3	orbit (4)	3.90 ±0.03	16.43 ±0.75	13.64 ±0.57	4.5	1.6	70
Augmented tree (base recipe)	PoE	1 view	4.85 ±0.03	17.33 ±0.31	20.15 ±0.57	4.2	3.3	64
Augmented tree (base recipe)	PoE	orbit (4)	3.87 ±0.03	12.30 ±0.50	12.25 ±0.37	3.0	1.7	279
+ orbit distillation	PoE	1 view	4.56 ±0.03	15.90 ±0.53	18.33 ±0.66	3.7	3.0	64
+ orbit distillation	PoE	orbit (4)	**3.69** ±0.02	**11.21** ±0.42	**11.09** ±0.49	2.6	1.4	279

Table 7. **SPEED+ rotation, ground-truth crops.** Mean E_R over 3 seeds (± seed std), fraction of severe flips (>150°), and end-to-end latency at batch 1. The clean tree is trained without augmentation. The DINOv3 student is distilled from a DINOv3-only orbit teacher, and the PoE students from the PoE orbit teacher. Inference cost is unchanged by distillation.

for DINOv3-only. It costs about 50 ms per view, half of which is the CPU consensus-mask step. DINOv3-only orbit voting runs at 14 FPS and a single DINOv3 view at 68 FPS (Appendix A).

Full 6-DoF. Table 19 (supplementary) adds translation with the SPACE-HOP target [5]. Because our crops come from ground-truth boxes, the box alone already locates the target to 0.55 m. The translation block improves on this by about 7–10× on synthetic data (both pre-registered criteria pass). Joint training does not hurt rotation and even helps slightly on real images: PoE + orbit reaches 3.86°/12.11°/11.73°, against 3.87°/12.30°/12.25°. Orbit averaging lowers translation error by 10–25%. The DINOv3 block is the most accurate on synthetic data (0.059 m) and reaches 0.185/0.183 m on the real domains; the MoGe-2 block alone is weak for translation. With a synthetic-only detector instead of ground-truth boxes, translation error rises to 0.66/0.27 m (last row of Tab. 6).

4.5. Adaptation and inference cost

Adaptation (Tab. 8). The backbones hold 634 M parameters (2.4 GB at fp32) and are trained for no application. What an application adds is small: a rotation tree of 2.1–2.4 M parameters per branch and a 0.33 M translation block, about 11 MB per object at fp32 with one branch and 20 MB with

Component	Status	Parameters	Storage
DINOv3 ViT-L/16	frozen, shared	303.1 M	1.13 GB
MoGe-2 ViT-L (optional)	frozen, shared	330.9 M	1.23 GB
Rotation tree, DINOv3 branch	trained per object	2.39 M	9.9 MB
Rotation tree, MoGe-2 branch	trained per object	2.13 M	8.4 MB
Translation block	trained per object	0.33 M	1.3 MB*

	Feature extraction (GPU-h, once)		Rotation tree (min/head)		Translation
Application	DINOv3	MoGe-2	DINOv3	MoGe-2	(min/head)
T-LESS	2.2	3.6	11	11	7
SYMSOL-T	2.6	3.4	7	9	6
YCB-V	2.6	4.3	7	5	9
SPEED+	2.2	3.9	13	6	16

Table 8. **What is shared and what is trained.** Top: parameters and fp32 storage of the frozen, shared backbones and of the per-object heads (median checkpoint size; * weights only). Bottom: one-off feature extraction per application (GPU-hours over all images and views, from the extraction logs) and median wall-clock training time per head on cached features (one shared GPU; upper bounds). The reported SPEED+ system uses distilled students (median 40 min per DINOv3 and 84 min per MoGe-2 head, after a one-off teacher pass) and the jointly trained DINOv3 6-DoF head (25 min); SPEED+ additionally needs a detector for deployment (6.3 h training, Sec. 4.4).

both, so one copy of the backbones can serve many objects. A head trains in a median of 5–16 minutes on one GPU, because features are extracted once per image and view; this extraction (2.2–4.3 GPU-hours per backbone and application) is the dominant one-off cost, and it scales with the number of training images and augmented views, not with the number of heads. The system as a whole is therefore not small; the part that has to be trained, stored and versioned per object is.

Inference (Tab. 9). At batch 1 the single-view DINOv3 system runs at 68 images per second (14.7 ms). In this regime the autoregressive tree costs about as much as the backbone (7.4 vs. 6.4 ms), whereas at batch 64 it drops to 0.1 ms per image. Orbit aggregation multiplies the end-to-end cost by about 4.7 (DINOv3) and 4.4 (PoE). The geometry branch is the expensive option: MoGe-2 and the CPU consensus mask add about 41 ms per view, so the fused single-view system runs at 15.6 images per second, and with orbit aggregation at 3.6. This is the accuracy–compute trade-off behind the per-application choices of Tab. 1: fusion is worth its $4\times$ cost on SPEED+ (lightbox $17.4^\circ \rightarrow 12.3^\circ$) but not on SYMSOL-T or YCB-V, where it does not help. Head-only latency and accuracy under reduced precision are measured in a pre-registered experiment that has not yet run (Appendix J.4).

5. Ablations

All ablations use the PoE variant, 3 seeds and the same evaluation run within each table. The training-target ablation runs on the general-object benchmarks (T-LESS, SYMSOL-T). The ablations of the head architecture, the orbit scoring rule and the augmentation were run on the spacecraft application (SPEED+), where the synthetic-to-real shift makes their effects largest; we do not assume that their magnitudes carry over to the other benchmarks.

Overview (Tab. 10). The table isolates one component per row. Multi-path targets and the translation block help on every benchmark where they apply, at no or negligible inference cost. Orbit aggregation helps on T-LESS, SYMSOL-T and SPEED+ at about 4–$5\times$ the inference cost, but hurts on YCB-V, so it is an option selected on validation. Geometry fusion is the most application-dependent component: it gives the largest single gain on the nearly symmetric spacecraft, a small gain on T-LESS, and none on SYMSOL-T and YCB-V, at about $4\times$ the cost. Distillation is evaluated only on SPEED+ so far. For the shared architecture this means that the core (frozen DINOv3, hierarchical head, multi-path targets, translation block) transfers across applications, while the expensive options have to be chosen per application.

Symmetry-aware targets. Table 11 compares the three training targets. Canonical labels (A1) remove the contradiction between equivalent labels and improve single-view accuracy on both datasets (SYMSOL-T $3.87^\circ \rightarrow 3.40^\circ$, T-LESS validation $12.3^\circ \rightarrow 11.5^\circ$). However, they break orbit voting: a canonical label is not related across in-plane rotations by Eq. (2), so the views disagree and the SYMSOL-T orbit error rises to 9.7°. Multi-path targets (A2) keep the relation and are the best single-view and orbit models on SYMSOL-T (2.72° and 2.10°; pre-registered criterion passes by about $100\times$ the seed std).

Hierarchical tree vs. flat grid (SPEED+). We replace the tree with a count-matched flat classifier over 131,076 Hopf-fibration anchors [27] in SPACE-HOP's configuration (10,923 directions $\times$ 12 rolls), keeping features, augmentation, training budget and orbit inference identical (Tab. 12). The tree is far more accurate (3.87° vs. 7.53° synthetic with orbit voting), better on real data in all comparisons, and uses $15\times$ fewer parameters per branch (2.4 M vs. 36.1 M, of which 33.6 M is the 131k-way output layer). Orbit voting helps both heads, and the flat head with orbit voting even has slightly fewer severe flips on lightbox, so flip repair comes mainly from orbit voting.

This comparison does not isolate hierarchical prediction. With only 12 in-plane rolls, the SPACE-HOP grid alone has

Configuration	Batch	DINOv3	MoGe-2 + mask	Head†	Total (ms/img)	FPS	Peak GPU (GB)
DINOv3, 1 view	1	6.4	0.0	7.4	14.7	68.1	2.4
	64	2.4	0.0	0.1	2.9	346.8	3.1
DINOv3, orbit (4)	1	26.3	0.0	40.0	69.6	14.4	2.4
	64	9.6	0.0	1.0	12.1	82.4	3.1
PoE, 1 view	1	6.9	41.2	14.1	64.0	15.6	2.8
	64	2.4	45.3	0.3	50.0	20.0	28.2
PoE, orbit (4)	1	28.1	164.9	76.4	278.5	3.6	2.8
	64	9.9	185.5	2.7	206.8	4.8	28.2

Table 9. **End-to-end inference cost** (ms per image) of the four configurations on SPEED+ crops, bf16, one shared NVIDIA RTX 6000 Ada, images pre-decoded. Total includes the crop warp(s); DINOv3 and MoGe-2 include feature pooling; the MoGe-2 column includes the CPU consensus-mask step; † head: rotation tree(s), orbit candidate generation and scoring. Not included: object detection and, for T-LESS and YCB-V, the roll-homography warp. Peak GPU memory is allocated memory of the whole pipeline.

Component	Benchmark	Matched comparison	Metric	Without	With	Extra cost
Multi-path targets	T-LESS	A0 vs. A2, PoE + orbit	test R@10 (%)	73.5	79.0	none
	SYMSOL-T	A0 vs. A2, DINOv3 + orbit	test E_R	2.62	2.07	none
Canonical labels	SYMSOL-T	A0 vs. A1, PoE + orbit	val E_R	2.65	9.73	none
Orbit voting	T-LESS	A2 PoE, 1 view vs. orbit	val E_R	10.82	8.28	4.4×
	SYMSOL-T	A2 DINOv3, 1 view vs. orbit	val E_R	2.54	2.07	4.7×
	YCB-V	DINOv3, 1 view vs. orbit	val E_R	8.97	11.53	4.7×
	SPEED+	aug. tree PoE, 1 view vs. orbit	LB / SL E_R	17.3 / 20.2	12.3 / 12.3	4.4×
Distillation	SPEED+	tree vs. student, PoE + orbit (vote)	LB / SL E_R	12.58 / 12.88	11.35 / 11.91	none
Geometry fusion (PoE)	SPEED+	DINOv3 vs. PoE, orbit	LB E_R	17.4	12.3	4.0×
	T-LESS	DINOv3 vs. PoE, A2 orbit	val E_R	8.61	8.28	4.0×
	SYMSOL-T	DINOv3 vs. PoE, A2 orbit	val E_R	2.07	2.10	4.0×
	YCB-V	DINOv3 vs. PoE, 1 view	val E_R	8.97	10.52	4.4×
Synthetic training data	YCB-V	real only vs. + renders, val-selected	test E_R	69.4	12.9	none
Hierarchical tree¶	SPEED+	flat 10,923×12 Hopf vs. tree, PoE + orbit	synth. E_R	7.53	3.87	–
Translation block	T-LESS	box only vs. block (orbit mean)	test E_T (cm)	6.3	2.6	+0.33 M
	YCB-V	box only vs. block (1 view)	test E_T (cm)	8.1	1.6	+0.33 M

Table 10. **Component contributions under matched controls.** Each row changes one component and keeps features, recipe, seeds and evaluation fixed; values are means over 3 seeds (° unless stated; lower is better except recall). Extra cost: end-to-end inference time relative to the configuration without the component, measured on SPEED+ crops (Tab. 9; the roll-homography warp used for T-LESS and YCB-V is not timed), or added parameters. ¶ the flat grid has a 6.06° quantisation floor against 2.51° for the tree, so this row does not isolate hierarchy; the same-grid control is pending (Tab. 12). Source runs are recorded next to each row in the table generator.

		T-LESS (7 obj.)		SYMSOL-T (5)	
Training target	Inference	val	test	val	test
single annotated label	1 view	12.3 ±0.1	17.9 ±0.2	3.87 ±0.03	3.88 ±0.01
	orbit (4)	9.3 ±0.1	16.6 ±0.2	2.65 ±0.01	2.67 ±0.00
canonical label	1 view	11.5 ±0.0	16.7 ±0.1	3.40 ±0.01	3.39 ±0.01
	orbit (4)	10.7 ±0.0	18.2 ±0.7	9.73 ±0.05	9.69 ±0.05
all equivalents (ours)	1 view	10.8 ±0.1	15.7 ±0.2	2.72 ±0.01	2.71 ±0.01
	orbit (4)	8.3 ±0.0	15.2 ±0.3	2.10 ±0.01	2.11 ±0.01
+ orbit distillation	1 view				
	orbit (4)				

Table 11. **Training targets for symmetric objects** (symmetry-aware mean error, PoE). Single annotated label (A0), canonical label closest to identity (A1), and all equivalent rotations (A2, Eq. (1)); the last block adds orbit distillation on top of the selected target. Blank cells: runs not yet completed.

Head (params/branch)	Grid	Floor	Inference	Synth.	LB	SL
Tesseract tree (2.4 M)	131,072 leaves	2.51	1 view	4.85 ±0.03	17.33 ±0.31	20.15 ±0.57
			orbit	3.87 ±0.03	12.30 ±0.50	12.25 ±0.37
Flat (36.1 M)	Hopf 10,923×12	6.06	1 view	8.95 ±0.06	20.56 ±0.70	27.16 ±1.04
			orbit	7.53 ±0.01	13.15 ±0.37	16.31 ±0.59
Flat, same grid	131,072 leaves	2.51	1 view			
			orbit			
Flat, balanced Hopf	Hopf 1,771×74	2.29	1 view			
			orbit			

Table 12. **Tree vs. flat grid** (PoE, mean error in °). Same frozen features, augmentation, training budget and orbit inference (4 views, top-8 per view). Floor: mean error of the nearest anchor (tree: its leaf) on the synthetic-validation rotations. The *same-grid* flat head classifies the tree's own 131,072 leaves, isolating hierarchical prediction; the balanced Hopf grid tests an alternative discretisation of the same size. Blank cells: runs not yet completed.

a quantisation floor of 6.06° on the synthetic-validation rotations, against 2.51° for the tree's leaves, so part of the gap

Selection over 32 candidates	Synth.	LB	SL
1 view (greedy)	4.85 ±0.03	17.33 ±0.31	20.15 ±0.57
max-confidence	5.51 ±0.03	14.63 ±0.42	15.92 ±0.15
top-1 vote	4.37 ±0.03	13.24 ±0.42	13.38 ±0.57
weighted vote (32)	3.46 ±0.03	12.58 ±0.42	12.88 ±0.49
orbit product	3.87 ±0.03	12.30 ±0.50	12.25 ±0.37

Table 13. **Selection rule over the same 32 orbit candidates.**

Augmentation	1 view			orbit (4)		
	Syn.	LB	SL	Syn.	LB	SL
none	5.54 ±0.07	26.90	36.66	4.62 ±0.04	20.25	24.83
photometric	5.00 ±0.03	17.38	19.68	4.25 ±0.06	12.29	12.37
in-plane roll	4.91 ±0.00	24.30	31.52	3.78 ±0.03	18.69	20.70
both	4.85 ±0.03	17.33	20.15	3.87 ±0.03	12.30	12.25

Table 14. **Which augmentation matters.** *None* is the clean tree. The roll uses exact labels (Eq. (2)).

is due to the grid. We therefore run a controlled comparison with two further flat heads, trained with the identical recipe and evaluated with identical orbit inference (4 views, top-8 candidates per view, orbit-product scoring). The first classifies the tree's *own* 131,072 leaf centres, and scores an orbit candidate by the probability of its Tesseract cell, exactly as the tree does; only the prediction (hierarchical vs. flat) differs. The second classifies a balanced Hopf grid of 1,771 Fibonacci directions × 74 rolls (131,054 anchors), an alternative discretisation of the same size whose floor (2.29° mean, 3.8° max) is *lower* than the tree's (2.51°, 5.8°). Both flat heads have about 33.6 M output-layer parameters per branch. Their results will fill the blank rows of Tab. 12; if the same-grid flat head matches the tree, the advantage in the first comparison is due to the grid rather than to hierarchical prediction.

Cost of the output layer. Measured on the prediction head alone with an earlier version of the architecture, a flat 131k-way head is about 3× faster than the tree's greedy decoding on a large GPU at batch 1 (0.40 vs. 1.23 ms), but 1.5× slower at batch 256, 2.7× slower on a small GPU and 7× slower on one CPU thread. In the full pipeline at batch 1, the autoregressive tree costs about as much as the DINOv3 backbone (7.4 vs. 6.4 ms, Tab. 9), while at batch 64 it costs 0.1 ms per image. The tree's established advantages are therefore its size and its behaviour on constrained hardware, not GPU latency at batch 1. The controlled comparison above also measures head and full-pipeline latency and training and inference memory for all three heads under identical conditions.

How to select among orbit candidates (SPEED+). With the same 32 candidates (Tab. 13), picking the most confident one is clearly worse than any form of agreement, and voting among the four views' top-1 predictions is better. Using all candidates helps most. The kernel-weighted vote is best on synthetic data (3.46°). The likelihood product of Eq. (3) is best on both real domains, but its advantage over the vote is within about one seed std. Its pre-registered criterion ("product beats voting on synthetic validation") *fails*, and under our selection protocol the weighted vote would be preferred. The product is reported in Tab. 7 only because it was fixed in the base recipe beforehand. We therefore claim multi-view consistency over hierarchical candidates, not a specific scoring rule.

Photometric vs. rotation augmentation (SPEED+). Table 14 separates the two augmentation families. Photometric augmentation with orbit voting reaches 12.3°/12.4° on real images, the same as the combined recipe. Rotation augmentation alone gives the best synthetic precision (3.78°) but leaves real errors at 18.7°/20.7°. Orbit voting on the clean tree already removes a quarter to a third of the real error (26.9° → 20.3°, 36.7° → 24.8°). The real-domain gain therefore comes from *lighting robustness plus orbit consistency*. The exact roll transform is needed for orbit voting itself; as a training augmentation it mainly improves synthetic precision. This corrects our initial hypothesis that rotation augmentation causes the flip repair.

Can training learn orbit consistency (SPEED+). On SPEED+ we compared paired views of the same image (a data control), a transported consistency loss on the leaf distribution (a standard equivariance regulariser), a *hierarchical* transported loss at every tree level, and orbit distillation (Tab. 20, supplementary). The hierarchical loss beats the base recipe and the paired-view control but not the leaf-only loss; its pre-registered criterion fails, so we make no hierarchical-transport claim. Both consistency losses mostly improve synthetic precision. Orbit distillation is the only objective that clearly improves the real domains (11.2°/11.1° with orbit voting).

Further ablations in the supplementary material cover the decoder lineage (MLP, GRU and Transformer decoders, hard vs. soft targets, beam search), fusion (single branches, learned fusion, weighted PoE), finer tokens, continuous flow posteriors and verifiers, discrete flow matching, oracle headroom, zero-shot transfer and timing.

6. Limitations

- **Ground-truth crops.** All benchmark results except the SPEED+ detector row use ground-truth boxes. On

SPEED+, a synthetic-only detector more than doubles the rotation error on lightbox ($11.35° \rightarrow 24.96°$) and raises the score from 0.229 to 0.541, mainly through poor real-domain boxes and the box-relative translation target (Tab. 6). A deployable system therefore needs a detector that transfers to real images, or translation that does not depend on the box. Detection was not evaluated on the general-object benchmarks.

- **Ground-truth translation in rotation inference.** On T-LESS and YCB-V the allocentric rotation is converted, and orbit candidates are mapped between views, with the ground-truth translation. The pre-registered evaluation with predicted translation, which also gives joint 6D success, has not yet run, so these rotation results are an upper reference for a system without ground-truth information.
- **How general the shared architecture is.** The backbones, feature interface, head architecture, recipe and inference algorithm are shared, but each object needs its own labelled training images, intrinsics, crops and symmetry set, and its own heads; heads do not transfer zero-shot to new objects (Appendix H). Two options are chosen per application on validation data: orbit aggregation helps on T-LESS, SYMSOL-T and SPEED+ but hurts on YCB-V, and MoGe-2 fusion helps only on SPEED+ and T-LESS. YCB-V also needed synthetic training data. Four applications are evidence for, not proof of, the claim that frozen backbones with small heads serve many applications; the industrial, warehouse and robotic settings named in the introduction were not evaluated.
- **Matched comparisons.** Published numbers use different crops, masks, object sets and fine-tuned backbones; Tabs. 2, 3, 5 and 6 state each setting, and only rows with the same setting are matched.
- **Hierarchy vs. grid.** The flat-baseline comparison of Sec. 5 uses a coarser grid than the tree; the same-grid control is still running, so the share of the gain due to hierarchical prediction is not yet established.
- **Quantisation floor.** The depth-5 leaf quantisation ($2.51°$ mean) bounds fine accuracy for asymmetric objects. Our best synthetic SPEED+ errors (3.3–$3.7°$) are close to it, and it caps fine-threshold recall on T-LESS. A continuous refinement on top of the selected mode is a natural next step; the refiners we tried helped only on synthetic data (Appendix C).
- **Cost.** The heads are small, but the system is not: the two backbones hold 634 M parameters, feature extraction is the dominant adaptation cost, orbit aggregation multiplies inference cost by about 4.5, and the geometry branch by about 4. Head-only latency and reduced-precision accuracy are not yet measured. Timings come from a shared GPU and are upper bounds.
- **Distillation** has so far been evaluated only on SPEED+; its cross-dataset evaluation is running.

7. Conclusion

Shared frozen appearance and geometry backbones can support 6D pose estimation of very different known objects through compact per-object heads. The same architecture, trained only in its 2.1–2.4 M-parameter rotation trees and 0.33 M-parameter translation block, reaches a recall of 79% at $10°$ and 88% at 5 cm on real T-LESS images, $2.07°$ on SYMSOL-T, $12.9°$ and 1.6 cm on YCB-V, and $11.3°/11.9°$ on real spacecraft images after training on renders only. The decisive problem for such heads is selecting the right mode among symmetric or look-alike rotations. Multi-path targets address it for exact symmetries on every benchmark at no inference cost, and orbit aggregation and distillation address look-alike flips where aggregation helps, at a measured 4–$5\times$ inference cost. The shared design has clear boundaries: the expensive options (geometry fusion, aggregation) pay off only on some applications, cluttered household objects need synthetic training data, heads do not transfer zero-shot, and deployment on real spacecraft images is currently limited by detection rather than by the heads. Fine-tuned state-of-the-art methods remain more accurate on every benchmark we use. We report validation-selected results, matched comparisons and negative results, so that the cost and the limits of adapting frozen backbones with small heads can be judged directly.

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Selecting the Right Mode: Symmetry-Aware, Orbit-Consistent Tesseract Trees on Frozen Foundation Features for 6D Object Pose Estimation

Supplementary Material

A. Timing

System	batch 1		batch 64	
	ms	FPS	ms/img	img/s
DINOv3, 1 view	14.7	68.1	2.9	347
DINOv3 + orbit	69.6	14.4	12.1	82
PoE, 1 view	64.0	15.6	50.0	20
PoE + orbit	278.5	3.6	206.8	5

Table 15. **End-to-end latency** of the augmented trees (shared RTX 6000 Ada; upper bounds; image already decoded).

At batch 1 (Tab. 15), a DINOv3 forward pass takes 6.4 ms and one tree 7.4 ms. MoGe-2 adds 21 ms for the network and 20 ms for the CPU consensus-mask step per view, which is the main optimisation target. Orbit voting multiplies the cost by 4.4–4.7×. Batched evaluation reduces tree cost to 0.1–0.3 ms per image.

B. Model lineage and fusion

Design (clean trees, PoE greedy)	Synth.	LB	SL
MLP path decoder, hard targets	6.24	31.1	43.7
GRU path decoder	6.15	–	40.6
MLP decoder, geodesic soft targets	6.07	28.2	42.1
Transformer decoder, hard targets	5.69	31.6	41.8
Transformer + soft targets	**5.54**	**26.9**	**36.7**
+ beam 8 instead of greedy	no gain	–	–
+ tangent residual at the leaf	−0.13	±0.0	±0.0
16×16 instead of 8×8 DINOv3 tokens	5.56	27.0	36.0
DINOv3 tree alone	5.35	37.1	–
MoGe-2 tree alone	–	43.7	–
Learned concat fusion, one Transformer (8×8)	5.34	40.8	–
Equal per-level PoE (same grid)	5.58	27.9	–
Weighted PoE, w_{DINO}=0.8 (proxy-selected)	5.14	32.0	36.7

Table 16. **Design steps** on clean (non-augmented) trees, seed means. Dashes: not evaluated in that study.

Table 16 lists the design steps that led to the base tree. The Transformer decoder with geodesic soft targets was best; beam search at full data brought no gain, and a tangent residual only refined errors within the correct cell. Fusion is strongly asymmetric across domains. A learned fusion that feeds concatenated tokens into one Transformer matches DINOv3 alone on synthetic data but is much worse on real images (40.8° vs. 27.9° for PoE on the same grid). We interpret this as the learned fusion over-fitting render statistics, while training-free PoE keeps the branches' errors partly independent. A weighted PoE selected on the rotated validation proxy (w=0.8) is 5° worse on lightbox than equal weights. Following our protocol, we did not use real data to choose another weight. The consensus foreground mask reduces MoGe-2's background leakage on real images from 0.52 to 0.12 (lightbox) and from 0.57 to 0.12 (sunlamp). It is what makes the MoGe-2 branch useful.

C. Negative results

Approach (on clean trees unless noted)	Synth.	LB	SL
Clean PoE tree (reference)	5.54	26.9	36.7
$SO(3)$ flow posterior on tree memory, KDE mode	5.22	29.8	37.6
best of 32 samples (*oracle*)	2.60	9.7	10.6
Flow re-noise verifier over beams + flips + corners	4.78	26.5	32.7
best candidate (*oracle*)	2.25	7.9	11.4
Kurz corner tree [15]	5.20	28.8	39.1
Tree distilled from flow masses (β=1, 1 seed)	5.62	26.2	37.7
Discrete flow matching on tree paths	7.22	29.8	39.8
Aug. DINOv3 tree (reference)	4.84	22.6	22.1
+ $SO(3)$ flow, KDE	4.10	25.2	24.4

Table 17. **Refiners and verifiers that did not fix real flips** (seed means). Oracles are not deployable and show headroom only.

Table 17 collects approaches that improved synthetic accuracy but not, or only slightly, real accuracy. A Riemannian flow-matching posterior [2] conditioned on the tree's encoder memory contains good samples: its best-of-32 oracle reaches 9.7° on lightbox. It also assigns the ground truth a much higher log-likelihood than the tree does (3.85 vs. 0.98 nats on lightbox), and its 90% regions over-cover (0.98). Yet its mode estimate is worse than the tree's. Re-scoring an expanded candidate set (beam, 180° flips, cell corners) by re-noise/return consistency helps synthetic data and sunlamp, but does not repair lightbox flips. Synthetic render-and-compare templates also failed to choose the right flip. A discrete flow on the tesseract path [6] was clearly worse. We note that interpolating path tokens is not geometry-preserving: 45–53% of intermediate paths lie more than 30° from both endpoints. Distilling flow masses into the tree did not pass its criterion. An oracle that stops decoding at the best depth would gain only 0.51° on synthetic data, so adaptive depth was not pursued. These results support the diagnosis of Sec. 4.4: the right mode is reachable, but scorers trained on clean synthetic features cannot select it on real images. Lighting-robust training and multi-

view consistency can.

D. Flip transitions on SPEED+

System	Lightbox		Sunlamp	
	fixed	broken	fixed	broken
Aug. tree, 1 view	51% of 401	1.0%	59% of 206	1.0%
Aug. tree + orbit	62% of 401	0.4%	80% of 206	0.4%
Aug. DINOv3 + orbit	52% of 401	1.3%	74% of 206	0.5%

Table 18. **Flip transitions relative to the clean PoE tree.** *Fixed*: share of the clean tree's $>150^\circ$ errors that become $<20^\circ$ (denominator given). *Broken*: share of its $<20^\circ$ predictions that become $>150^\circ$.

E. Orbit distillation and 6-DoF details

Arm	Feat.	$\hat{t}$	E_T (m)			E_R (°), orbit		
			Syn.	LB	SL	Syn.	LB	SL
Box only	–	–	0.561	0.547	0.553	–	–	–
Frozen	DINOv3	1 view	0.090	0.233	0.240	4.21	17.40	15.27
		orbit	0.068	0.205	0.205			
Frozen	PoE	1 view	0.108	0.206	0.232	3.87	12.30	12.25
		orbit	0.089	0.184	0.206			
Joint	DINOv3	1 view	0.078	0.209	0.212	4.19	17.31	14.87
		orbit	**0.059**	0.185	**0.183**			
Joint	PoE	1 view	0.111	0.196	0.221	3.86	12.11	11.73
		orbit	0.093	**0.175**	0.194			

Table 19. **SPEED+ 6-DoF** with the SPACE-HOP translation target, ground-truth crops. Rotation uses PoE or DINOv3 orbit voting. Translation ($\hat{t}$) is predicted from one view or as the orbit average; seed std of $E_T \leq 0.005$ m. *Box only* uses no image: the object centre is placed at the box centre and the depth at the training mean of U_z.

Training objective	1 view, DINOv3		PoE + orbit		
	Syn.	Syn.-rot.	Syn.	LB	SL
Base recipe (GT loss)	4.84 ±0.05	9.88 ±0.22	3.87	12.30	12.25
+ paired views	4.96 ±0.06	10.04 ±0.21	3.94	12.38	12.45
+ leaf-level transported consistency	4.59 ±0.04	9.43 ±0.14	3.71	12.41	11.74
+ hierarchical transported consistency	4.65 ±0.03	9.57 ±0.06	3.74	12.49	11.73
+ orbit distillation	4.57 ±0.05	9.33 ±0.18	**3.69**	**11.21**	**11.09**

Table 20. **Training-time consistency.** Single-view DINOv3 trees on synthetic and rotated synthetic validation (left) and PoE + orbit (right). The consistency losses transport each view's beam distribution into another view's frame via Eq. (2).

Retention of the orbit gain by the single-view students, defined as (base greedy − student greedy)/(base greedy − base orbit), is 43%/18%/22% (DINOv3) and 30%/28%/23% (PoE) on synthetic/lightbox/sunlamp, and 13–15% on the rotated proxy. For the joint 6-DoF model, the ESA-style score (rotation error in radians plus normalised translation error, with the challenge precision floors) is 0.084/0.241/0.238 for PoE.

F. Per-object results

Object	PoE+orbit	DINOv3+orbit
T-LESS		
1	23.82	25.00
4	14.50	15.69
5	6.22	7.07
11	11.90	12.92
18	14.38	14.78
21	34.78	37.24
27	10.36	12.43
SYMSOL-T		
cone	1.83	1.79
cube	2.72	2.90
cyl	1.67	1.65
icosa	3.81	3.71
tet	3.35	3.06
YCB-V		
003_cracker_box	85.27	74.61
006_mustard_bottle	97.96	87.29
011_banana	61.63	83.23
025_mug	24.25	22.76
035_power_drill	41.34	25.07
037_scissors	169.07	123.72

Table 21. **Per-object test error** of the single-label (A0) trees (3-seed mean; symmetry-aware for T-LESS and SYMSOL-T; YCB-V trained on real videos only).

On T-LESS (Tab. 21), most of the test error comes from object 21 (no symmetry metadata, near-symmetric; 18% of predictions above 90°) and object 1 (continuous symmetry). Continuous symmetries are evaluated on a 10°-step discretisation of the symmetry set, which adds up to 5° of artificial error for objects 1 and 4 and for the cone and cylinder.

G. YCB-V diagnosis

On YCB-V we checked four things:

1. The trees fit their training frames: 3.8° and 3.9° on mustard bottle and mug.
2. Test poses are covered by training: the median nearest training rotation is about 5° away for cracker box, mustard bottle and banana, whose test errors are still 75–87°.
3. Every crop contains all of the object's label pixels.
4. The ground-truth rotations agree with the segmentation: 65–98% of the projected model points fall on the object mask, against 28–83% for a transposed-rotation control.

The failure is therefore memorisation of the 16–20 cluttered training videos per object. The object covers only 5–40% of the crop and the DINOv3 tokens are not masked. This

is not a label or pipeline error. Standard YCB-V training adds the synthetic `data_syn` renders, whose backgrounds are black. We pasted each render onto a real frame drawn only from the training sequences (never validation or test), capped at 12,000 crops per object as for the real data, and retrained with the unchanged recipe. Table 22 shows the effect per object.

	real videos		+ synthetic	
Object	DINOv3	PoE	DINOv3	PoE
003_cracker_box	77.7	78.7	5.1	5.7
006_mustard_bottle	101.5	102.4	6.1	7.6
011_banana	79.9	77.6	6.1	5.3
025_mug	24.6	28.4	6.5	5.6
035_power_drill	30.3	38.2	4.3	4.4
037_scissors	113.2	161.5	49.4	36.9

Table 22. **YCB-V per-object test error** (°, single view, 3 seeds) without and with synthetic training data.

H. Zero-shot transfer of the frozen SPEED+ trees

Dataset	Raw E_R		Frame-free pair error		
	frozen	chance	frozen	chance	constant
T-LESS	117.5	103.6	95.4 $\pm$3.0	104.5	88.7
SYMSOL-T	49.4	53.7	47.8 $\pm$0.5	53.7	53.8
YCB-V	115.6	125.8	101.6 $\pm$2.8	125.8	57.8

Table 23. **Zero-shot**: frozen SPEED+ PoE trees with orbit voting, no retraining, test split. The pair error $\min_S \angle(\hat{R}_i \hat{R}_j^\top, R_i S R_j^\top)$ over random same-object pairs cancels the unknown offset between object frames. *Chance*: uniformly random predictions. *Constant*: one rotation predicted for every image.

Object frames are arbitrary, so raw errors are expected near chance. The frame-free pair error (Tab. 23) is therefore the meaningful measure. On T-LESS and YCB-V it is *worse* than predicting one constant rotation, because both datasets have strong pose priors. On SYMSOL-T, where rotations are uniform, the trees beat chance only because of one shape: the cube (27.8° vs. 41.2°), plausibly because the spacecraft body is box-like. There is no general zero-shot transfer: the trees must be retrained per object.

I. Disclosures and reproducibility

- **Late feature-equivalence check.** The augmented and rotated views are extracted by a compact feature extractor. A pre-registered check against the full feature caches was run after the main results. It failed its strict element-wise tolerance: the maximum standardised difference was 0.51 for MoGe-2 features (p99 0.004; correlation 0.999998). Its functional follow-up showed that no PoE prediction changed on 512 crops, and that single-branch greedy leaves changed for 1/512 (DINOv3) and 3/512 (MoGe-2) crops.
- **SYMSOL-T bug.** Nine MoGe-2 SYMSOL-T trees first trained to NaN: always-empty border cells had a standard deviation near 10^{-6}, and float16 overflowed. They were retrained with a standard-deviation floor of 10^{-2}; all other trees were unaffected. Only the corrected evaluation is reported.
- **YCB-V paths.** A file-path error in the YCB-V manifest (colour images are JPEG) was fixed before any YCB-V result.
- **Post-hoc control.** The constant-prediction control in Tab. 23 was added after the T-LESS zero-shot result and is labelled as post hoc.

Every number in the results tables is generated by a script directly from the evaluation logs of the named runs.

J. Project details

This appendix summarises the whole project in one place: how the method works in detail, how data, training and evaluation were set up, what every experiment behind the paper found, how the design was arrived at, what failed, and what we do and do not claim. Unless stated otherwise, errors are mean geodesic rotation errors in degrees, averaged over 3 seeds, and $\pm$ is the sample standard deviation over seeds. "Synth" is the SPEED+ synthetic validation set, "val-aug" its augmented copy (Appendix J.2), and LB/SL are the lightbox and sunlamp test domains. Quaternions are scalar-first, $q = (w, x, y, z)$, and $\mathrm{Exp}(\phi\, d) = (\cos\frac{\phi}{2}, \sin\frac{\phi}{2}\, d)$ for a unit axis d.

J.1. Method details

Tesseract partition. A unit quaternion q is assigned to the chart $c = \arg\max_i |q_i|$ (first maximum on ties) and multiplied by $\mathrm{sign}(q_c)$, so q and $-q$ get the same chart and the same path. The quaternion is projected to the cube facet, $x = q/|q_c|$, and its three free coordinates $u \in [-1, 1]^3$ are halved recursively. At each level, each coordinate goes to the upper half ($b_i = 1$) if u_i is at least the interval midpoint, which gives the child id $k = b_0 + 2b_1 + 4b_2 \in \{0, \ldots, 7\}$. The leaf id at depth L is

$$\mathrm{id} = c \cdot 8^L + \textstyle\sum_{\ell=1}^{L} k_\ell\, 8^{L-\ell}, \tag{7}$$

so there are $4 \cdot 8^L$ leaves. After l decisions, free coordinate i of the cell centre is $u_i = -1 + \sum_{t=1}^{l} b_{t,i} 2^{1-t} + 2^{-l}$; the centre quaternion sets $x_c = 1$ and normalises. The tree uses cell centres only. This follows the recursive tesseract of Kurz *et al*. [15], but uses cube centres instead of corners (a corner has no unique parent), keeps only the 4 positive facets, and adds an explicit encoder, decoder and tie rule. The encoder invariants (antipodal invariance, encode–decode round trip, consistent prefixes) hold on all 69,491 SPEED+ labels.

Table 24 gives the quantisation by depth. The error halves with every level. The grid is not uniform: at L5 cell volumes differ by up to $14.4\times$, so a uniform prior over leaves is not uniform over rotations. We stop at $L = 5$: L6 would have 1,048,576 cells against 47,966 SPEED+ training images. Per hypothesis, the Tesseract is less precise than Super-Fibonacci and Hopf grids [27] of the same size (L5: 2.51° mean against 2.47° for Super-Fibonacci). Its advantage is search cost: greedy descent scores only 44 nodes at L5.

Table 24. **Tesseract quantisation by depth** ($2{\cdot}10^6$ Haar rotations). Mean: angle to the centre of the rotation's own cell (what the tree returns). Max: exact centre-to-corner bound.

L	cells	mean (°)	max (°)
1	32	39.27	81.79
2	256	19.95	46.83
3	2,048	10.01	24.43
4	16,384	5.01	12.36
5	131,072	2.51	6.20

Frozen features. Both backbones are frozen (evaluation mode, no gradients, asserted after every pass). Features are extracted once per crop and view into an fp16 cache, and the trees read only these caches.

- **DINOv3, `tokens8`.** DINOv3 ViT-L/16 [22] on a 256×256 crop with ImageNet normalisation. The last-block patch tokens ($16 \times 16 \times 1024$; CLS and register tokens dropped) are average-pooled to 8×8, giving 64×1024 tokens.
- **MoGe-2, `normals16~and`.** MoGe-2 [24] on a 518×518 crop predicts camera-frame normals n and a mask, both pooled to 74×74. The *consensus mask* m is the AND of the MoGe-2 mask (> 0.5) and a training-free DINOv3 foreground: the first principal component of the 256 patch tokens, sign-flipped so the border is background, upsampled and Otsu-thresholded. The tokens are $X = [\mathrm{pool}_{16}(n \odot m),\ \mathrm{pool}_{16}(m)] \in \mathbb{R}^{256\times 4}$. On real images the AND mask cuts the foreground outside the keypoint envelope from 0.52 to 0.12 (LB) and from 0.57 to 0.12 (SL) compared with MoGe-2's own mask.
- **Standardisation.** Every feature dimension is standardised as $(x - \mu)/(\mathrm{sd} + 10^{-6})$ with statistics from the clean training rows only (per object on the non-SPEED+ datasets), and the same statistics are applied to augmented, validation and test views. The MoGe-2 trees for three SYMSOL-T shapes use an sd floor of 10^{-2} (Appendix J.5).

Tree network. Each token is mapped by LayerNorm and a linear layer to $d = 256$, and a fixed 2-D sinusoidal encoding is added. Its frequencies are $\omega_k = 2^k\pi$ for $k = 0, \ldots, 63$, *i.e*. up to $2^{63}\pi$; only the lowest few ($2^k < 2g$ for a $g \times g$ grid) separate grid positions, and in float32 the higher channels hold fixed, position-dependent rounding residues. One pre-norm Transformer encoder layer produces the memory, and a causal decoder with 2 layers (4 heads, FFN 512, dropout 0.1) reads the sequence $(r, s_1, \ldots, s_L)$: a learned root token, then $s_{j+1} = E_{\mathrm{lev}}[j] + W\varphi(c_j)$, where c_j is the centre of the cell reached after j decisions and φ is the quaternion plus a 6-frequency Fourier encoding (52 dimensions). Output 0 feeds a 4-way chart head and output $j \geq 1$ an 8-way child head. Under teacher forcing one decoder pass gives all $1 + L$ distributions. The DINOv3 tree has 2,391,564 trainable parameters and the MoGe-2 tree 2,128,404.

Soft targets, beam search and PoE. With $\delta(q,c) = 2\arccos|\langle q,c\rangle|$ in degrees, the target at level ℓ is

$$t_k^{(\ell)} = \mathrm{softmax}_k\big(-(\delta(q,c_{\ell,k})/\tau_\ell)^2\big), \tag{8}$$

over the 4 chart centres ($\ell = 0$) or the 8 children of the *ground-truth* parent, so mass never leaves the true parent cell. The widths are fixed a priori to the mean cell error of each level: $\tau_0,\dots,\tau_5 = 39.27, 39.27, 19.95, 10.01, 5.01, 2.51°$. The loss $\mathcal{L}_{\mathrm{GT}}$ sums the six soft cross-entropies. Beam search keeps the B best prefixes by cumulative log-probability; greedy is $B = 1$, and beams up to 8 did not change the single-view result (PoE synth 6.242° at B=1, 6.248° at B=8). The two branches are fused at every node by the parameter-free product of experts [9],

$$\log\tilde{p}^{(\ell)} = \log\mathrm{softmax}\Big(\textstyle\sum_b \log\mathrm{softmax}(z_b^{(\ell)})\Big). \tag{9}$$

Augmentation and the exact roll. Every training crop has two pre-extracted augmented copies (seeds 1 and 2). Each copy draws a roll $\theta \sim \mathcal{U}(-180°, 180°)$ and a photometric chain: brightness, contrast, gamma, blur ($p = 0.5$), glare blobs ($p = 0.3$), Gaussian noise, random erasing ($p = 0.3$) and JPEG ($p = 0.5$). The rolled crop is resampled from the full image in one warp, so there are no black corners. A roll θ (counter-clockwise on screen) about the unit ray d through the crop centre changes the labels as

$$q' = \mathrm{Exp}(-\theta d)\otimes q, \qquad t' = \mathrm{Rot}(d,-\theta)\,t \tag{10}$$

(the "pre−" convention). It was derived from the camera geometry and checked before any training on 512 synth crops at 4 angles: keypoints agree to 0.34 px mean (1.05 px p95), the label error is 0.06° mean (1.14° max), and the other three conventions are off by 94–242 px. On SPEED+, d is computed from the *undistorted* crop centre. On T-LESS and YCB-V objects lie far from the optical axis and the affine warp failed the same rule (p95 label error 1.43° / 2.40°), so the crop is warped by the exact roll homography $C\,K\,\mathrm{Rot}(d,-\theta)\,K^{-1}$. The labels then follow exactly (error 0.000°), and the allocentric target is recomputed from t'. SYMSOL-T uses the full frame, whose centre is the principal point, so the affine roll is already exact. A roll multiplies on the left, so it commutes with right-multiplication by an object symmetry.

Orbit voting and scoring rules. At test time four views $\theta_k \in \{0, 90, 180, 270\}°$ are extracted. In each, the fused tree runs beam search with $B = 8$, and the 8 leaf centres are mapped back to view 0 by the inverse roll (on allocentric datasets via the egocentric frame, using the ground-truth translation). This gives 32 candidates. The *orbit product* maps each candidate c into every view, encodes it to its own depth-5 path, and sums the teacher-forced fused log-probabilities:

$$s(c) = \textstyle\sum_k \big[\log\tilde{p}_k^{(0)}(\text{chart}) + \sum_{\ell=1}^{5}\log\tilde{p}_k^{(\ell)}(k_\ell \mid \text{prefix})\big]. \tag{11}$$

The prediction is $\arg\max_c s(c)$. EXP-114 compared it with three simpler rules over the same candidates: *max-confidence* (highest beam score in its own view), *top-1 vote* (Gaussian-kernel support, $\sigma = 5°$, among the four view top-1s) and *weighted vote* (kernel support from all 32 candidates, weighted by the per-view softmax of beam scores). The product re-scoring costs 8.2 ms (DINOv3-only) or 15.3 ms (PoE) per image at batch 1; the weighted vote needs below 0.1 ms.

Symmetry-aware targets. For an object with symmetry set $\mathcal{S}$, three label modes were compared (EXP-106):

- **A0, single label**: the annotated rotation q_v of the view.
- **A1, canonical**: in each view separately, the equivalent $q_v \otimes s$ closest to the identity. Because it is per view, the labels of a view and its rolled copy need not be related by the roll transform.
- **A2, multi-path**: every equivalent $q_v \otimes s$ gets its own path and soft targets, and the loss is the mean of $\mathcal{L}_{\mathrm{GT}}$ over them (all of $\mathcal{S}$ if $|\mathcal{S}| \le 8$, otherwise 8 drawn per sample and step). For $|\mathcal{S}| = 1$ this is exactly A0.

The pre-registration named a hard-path level-marginal loss for A2. An amendment written before any run replaced it by the smoothed loss above, because the hard-path loss would not reduce to A0 for asymmetric objects.

Orbit distillation. The teacher is the frozen orbit system of the same seed and variant, applied to each training image over its three existing views (clean and two augmented copies, with known rolls). It yields 24 candidates c_j with weights $w_j = \mathrm{softmax}_j\, s(c_j)$, computed once and stored as constants. A new student tree is trained from scratch with

$$\mathcal{L} = \mathcal{L}_{\mathrm{GT}} - \beta\textstyle\sum_j w_j \frac{1}{L+1}\sum_{\ell=0}^{L}\log p^{(\ell)}\big(\mathrm{pre}_\ell(c_j)\big), \tag{12}$$

where $\log p^{(\ell)}(\mathrm{pre}_\ell(\cdot))$ is the joint log-probability of the length-ℓ prefix of the candidate's path, transported into the drawn view, and $\beta = 1$ was fixed before any run. Inference is unchanged. This is data distillation [8, 21] with a hierarchical target.

Translation block. We keep the SPACE-HOP target [5] unchanged. With the ground-truth square box (b_x, b_y, b_w, b_h), image size (W, H), $\alpha = 1.6$ and $\kappa = \frac{1}{2}(\alpha b_w/W + b_h/H)$, the target is $u = \big((f_x X/Z + c_x - b_x)/b_w,\ (f_y Y/Z + c_y - b_y)/b_h,\ \kappa Z\big)$. A learned query cross-attends once (4 heads) to the tree's

encoded tokens, followed by LN $\to$ Linear$(256, 256)$ $\to$ GELU $\to$ Linear$(256, 3)$: 331,011 parameters per branch. It is trained jointly with the tree ($\mathcal{L}_{\text{GT}} + \|u - u^*\|^2$; arm J) or on a frozen tree (arm F; also used on the other datasets, where u is standardised). PoE averages the two branches' u, and orbit inference averages the four back-rolled estimates. The *box-only* reference uses no image: object centre at the box centre, depth at the training mean.

Flat baselines. The flat classifier is the same network with path decoding removed: the decoder runs on the root token alone and feeds one linear layer over K anchors. For $K = 131{,}076$ the head alone has 33,686,532 parameters (36,078,096 in total for DINOv3, against 2.4 M for the tree). It is trained with geodesic soft targets over the 32 anchors nearest to the label at the tree's leaf width $2.51°$, with the tree's optimiser, views and checkpoint rule. In orbit voting it scores a candidate by the log-probability of its hard anchor. Three grids are used: SPACE-HOP's Hopf grid at the count-matched $10{,}923 \times 12 = 131{,}076$ (EXP-113), the Tesseract's own 131,072 leaf centres, and a balanced Hopf grid $1{,}771 \times 74 = 131{,}054$ (EXP-127).

J.2. Data, training and evaluation protocol

Benchmarks. Table 25 summarises the four benchmarks. All rotation results use ground-truth square crops: the tight box of projected object points, made square with side $\max(w, h, 32\,\text{px}) \cdot 1.3$ (15% margin per side). On SPEED+, the 11 Tango keypoints are used only to build these boxes and to verify conventions, and never as a training target. *Val-aug* is the 11,994 synth images with one fixed augmented copy (seed 1000). It was added in EXP-109 as a second selection domain, because clean synth has almost no gross errors and real images may not be used for selection. On T-LESS and YCB-V the tree predicts the allocentric rotation $R_{\text{ray}}(t)^\top R$, where $R_{\text{ray}}(t)$ is the minimal rotation taking the optical axis to the object ray.

T-LESS continuous symmetries are discretised in 36 steps of $10°$, so for objects 1 and 4 our symmetry-aware error can exceed the true one by up to $5°$. Its symmetry sets have sizes 36, 36, 2, 2, 1, 1 and 4; SYMSOL-T groups have 12 (tetrahedron), 24 (cube), 60 (icosahedron), 360 (cone) and 720 (cylinder) elements. The EXP-122 YCB-V composites paste `data_syn` renders onto backgrounds drawn only from real *training* frames.

Training. Table 26 lists the shared recipe. It was fixed for SPEED+ at the Phase-3 gate, locked by DEC-018, and not tuned per dataset. One tree is trained per backbone (and per object outside SPEED+).

Metrics.

- **Rotation error** $E_R = 2 \arccos |\langle \hat{q}, q \rangle|$. The *symmetry-aware* error (T-LESS, SYMSOL-T) is $\arccos\left(\frac{1}{2}(\max_{S \in \mathcal{S}} \text{tr}(\hat{R}^\top R S) - 1)\right)$; on T-LESS the raw error is 38–47°, which is why it is needed.
- **acc@k / R@k**: fraction with error below $k°$ (strict on SPEED+, $\le$ elsewhere).
- **Severe flips**: fraction with $E_R > 150°$. Flip transitions against a reference: *fixed* = share of its $> 150°$ errors that become $< 20°$, *broken* = share of its $< 20°$ predictions that become $> 150°$.
- **Translation**: $E_T = \|\hat{t} - t\|$, normalised $e_t = E_T / \|t\|$, and T@c = fraction with $E_T \le c$ cm.
- **Official SPEED+ score** [17, 19]: the mean over images of e_q (rad) $+ e_t$, with each term set to zero below the challenge floors ($0.169°$, 2.173×10^{-3}).
- **Retention** of distillation: (base greedy − student greedy) / (base greedy − base orbit).

Outside SPEED+, errors are first averaged per object and then over objects with equal weight.

Selection and pre-registration. The split policy (DEC-000, restored by DEC-006) is: train on synthetic/train, select on synthetic validation (and from EXP-109 on also on val-aug), and treat lightbox and sunlamp as test-only and descriptive. A seeded 20% lightbox split (DEC-005) was withdrawn the same day, before it influenced any run. The other benchmarks select on held-out scenes or sequences. Each experiment has a report that records, before any run, the question, arms, selection data, metrics and pass criterion. The standard pass rule compares 3-seed means on validation data: a difference passes only if it is larger than the larger of the two seed standard deviations. Protocol changes are dated amendments written before the affected runs; deviations found later are recorded as such. Real-domain and test numbers are reported for every arm and never used for selection.

Detected crops (EXP-124). The detected-crop evaluation was pre-registered before the detector was trained. An Ultralytics YOLO11s detector (COCO-pretrained) is trained on synthetic/train only, at image size 960 for 20 epochs, with the tight keypoint box as its single-class label and the checkpoint chosen on synthetic validation. At test time the top-1 box is turned into a crop with the ground-truth rule; when there is no detection the whole image is used, and these cases are counted. Systems S1–S3 were fixed before any result, and a ground-truth-crop run with the same code must first reproduce EXP-117 rotation and EXP-118 translation.

Compute. All logged runs used one NVIDIA RTX 6000 Ada in a workstation shared with other users, so timings are upper bounds. Feature extraction on SPEED+ took 11.5 ms

Table 25. **Benchmarks.** Counts are images (SPEED+) or object instances. All crops are ground-truth square boxes with 15% margin.

	Splits (train / val / test)	Crop	Target	Metrics
SPEED+	synthetic train 47,966 / synthetic val 11,994 (+ val-aug) / lightbox 6,740, sunlamp 2,791 (test only)	11 projected Tango keypoints, with distortion; gray → 3 channels	egocentric quaternion	raw E_R, acc@$\{1, 3, 5, 10, 20\}$ ($<$), severe flips, official score
T-LESS	7 objects (1, 4, 5, 11, 18, 21, 27): PBR scenes 0–44 + real, $\leq$12,000 per object, 84,000 / PBR scenes 45–49, 14,030 / BOP19 real targets, 2,345	projected 3-D model box corners; RGB	allocentric	symmetry-aware E_R, R@$\{2, 5, 10\}$ ($\leq$)
SYMSOL-T	5 shapes: 20,000 per shape (100,000 used) / 2,000 per shape / test-5k, 25,000; rendered locally with LiePose's generator	full 224×224 frame	egocentric	symmetry-aware E_R (annotated groups), R@k
YCB-V	6 objects (PoseCNN release): real 48,316 (+ 72,000 synthetic in EXP-122) / held-out sequences 0012, 0047, 0065, 0074: 2,797 / keyframes, 3,478	projected model points; RGB	allocentric	raw E_R (no symmetry metadata)

Table 26. **Training recipe** shared by all trainers.

Epochs, batch	100, 256
Optimiser	AdamW, lr 10^{-3}, weight decay 0.05
Schedule	OneCycle, 5% warm-up
Precision	bf16 autocast, fp16 features
Tree	$d = 256$, 1 encoder + 2 decoder layers, 4 heads, FFN 512, dropout 0.1
Views	clean or augmented copy 1 or 2, drawn uniformly per sample and step
Soft targets	τ_ℓ = mean cell error of level ℓ (flat: 32 anchors, 2.51°)
Distillation	$\beta = 1$, 24 teacher candidates
Checkpoint	validated every 5 epochs; lowest clean synth greedy mean (SPEED+) or symmetry-aware val greedy mean (others)
Seeds	0, 1, 2

per image for DINOv3 and 29.2 ms for MoGe-2. A locked-recipe SPEED+ tree trains in a median of 13.3 min (DINOv3) or 5.8 min (MoGe-2), and per-object trees on the other benchmarks in a median of 2.3–11.1 min. Every heavy job runs under a memory cap and an in-process memory guard, with a watchdog over the job queues.

J.3. Results summary

Table 27 lists every experiment behind the paper with its pre-registered outcome. Two evaluation passes of the same trees can differ in the second decimal (*e.g.* sunlamp PoE + orbit is 12.28 in EXP-109/112 and 12.25 in EXP-113–117); each row quotes its own run.

Orbit voting hurts on YCB-V. With synthetic training data, orbit voting makes YCB-V validation *worse* for both variants (PoE greedy 10.52 vs orbit 12.74; DINOv3 8.97 vs 11.53), and it does not help the test mean (PoE 10.90 vs 12.16). Test seed spread is large (DINOv3 greedy $\pm$6.90). The validation-selected YCB-V system is therefore DINOv3 single-view (test 12.92). On T-LESS real test images orbit voting helps the mean but removes few gross errors ($> 90°$ fraction 0.055 to 0.051).

Official score. On ground-truth crops the official SPEED+ score is 0.229/0.239 (LB/SL) for S1 with the validation-selected weighted vote and 0.227/0.224 with the pre-registered likelihood product (S1: the distilled PoE students with orbit aggregation plus the jointly trained DINOv3 translation block). The jointly trained 6-DoF PoE model (S2) scores 0.241/0.238. Applying the floor jointly instead of per component changes any per-seed domain score by at most 1.1×10^{-4}. The score is dominated by the rotation term (S1 on LB: 0.196 rotation, 0.031 translation).

Criteria that failed.

- **EXP-109 B**: 16×16 tokens give val-aug 30.51 vs 30.84, within the seed std, at $4\times$ the tokens.
- **EXP-114**: the orbit product does not beat the weighted vote on synth. The claim is therefore "multi-view orbit consistency over hierarchical candidates", not a specific rule; the product stays in the recipe only because DEC-018 fixed it beforehand.
- **EXP-116 criterion 1**: the hierarchical transport loss beats the base and the paired-view control but not a plain leaf-level equivariance loss, so no hierarchical-transport claim is made.
- **EXP-064 fusion on SYMSOL-T and YCB-V**: MoGe-2 adds nothing on the textureless SYMSOL-T renders and hurts on real-only YCB-V. The fusion claim is restricted to SPEED+ and T-LESS.

Distillation retention. The student recovers part of the orbit gain at single-view cost. Retention on synth / val-aug / LB / SL is 43% / 15% / 18% / 22% for DINOv3 and 30% /

Table 27. **Results summary.** Triples are synth / LB / SL; "val" is the validation split of the named dataset. Outcomes follow the larger-seed-std rule. EXP-106 and EXP-122 outcomes were evaluated from the run CSVs against the pre-registered rule.

Exp.	Question	Key numbers	Outcome
109	Do augmentation (A), 16×16 tokens (B) or orbit voting (C) help?	PoE val-aug: A 9.75 vs 30.84 (clean tree); B 30.51 vs 30.84; A+C 5.82. A+C 3.87 / 12.31 / 12.28 vs 5.54 / 26.91 / 36.66	A PASS, B FAIL, C PASS
112	Cost of the four locked systems	PoE+orbit 3.87 / 12.31 / 12.28 at 278.5 ms; DINOv3+orbit 4.21 / 17.40 / 15.27 at 69.6 ms; DINOv3 greedy 4.84 / 22.56 / 22.02 at 14.7 ms (batch 1)	descriptive
113	Tree vs flat classifier on SPACE-HOP's Hopf grid ($10{,}923 \times 12$)	PoE+orbit synth 3.87 vs 7.53, val-aug 5.82 vs 9.80; greedy 4.85 vs 8.95; LB 12.30 vs 13.15, SL 12.25 vs 16.31; 2.4 M vs 36.1 M parameters per branch	PASS (grid confounded, see 127)
114	Does the orbit product beat simpler rules?	synth: weighted vote 3.46 vs product 3.87 (PoE), 3.71 vs 4.21 (DINOv3); val-aug PoE 5.82 vs 5.98, DINOv3 6.25 vs 6.30; product best on real	FAIL on synth; marginal PASS on val-aug (PoE only)
115	Which augmentation matters?	PoE+orbit: both 3.87 / 12.30 / 12.25; photometric 4.25 / 12.29 / 12.37; roll 3.78 / 18.69 / 20.70; none 4.62 / 20.25 / 24.83	both criteria PASS (val-aug)
116	Roll-consistent hierarchical loss (T)	DINOv3 greedy val-aug: T 9.57, A 9.88, C1 10.04, C2 (leaf-only) 9.43; cons.@10 T 0.603 vs A 0.554	crit. 1 FAIL, crit. 2 PASS
117	Orbit distillation into a single-view student	student greedy val-aug DINOv3 9.33 vs 9.88, PoE 9.22 vs 9.75; student PoE+orbit 3.69 / 11.21 / 11.09	PASS
118	Translation block for 6-DoF	E_T synth: box only 0.561 m; joint DINOv3 orbit 0.059 / 0.185 / 0.183 m; joint PoE 0.093 / 0.175 / 0.194 m; PoE orbit E_R 3.86 vs 3.87	PASS (both criteria)
104	Cause of the $\sim 180°$ real errors	36–41% of $> 90°$ errors fall in 165–180° (20% if random); top-8 beam holds a path within 20° for 86% of LB; flat heads flip as often	analysis: visual near-symmetry
064	Recipe on T-LESS, SYMSOL-T, YCB-V	PoE vs DINOv3, val orbit: T-LESS 9.25 vs 9.94; SYMSOL-T 2.65 vs 2.61; YCB-V 74.06 vs 62.99. Test: T-LESS PoE+orbit 16.57, SYMSOL-T DINOv3+orbit 2.62	fusion: T-LESS PASS, SYMSOL-T FAIL, YCB-V FAIL; zero-shot: no claim
106	Symmetry targets A0 / A1 / A2	PoE greedy val: T-LESS 12.30 / 11.46 / 10.82; SYMSOL-T 3.87 / 3.40 / 2.72. PoE+orbit test: T-LESS 16.57 / 18.24 / 15.19; SYMSOL-T 2.67 / 9.69 / 2.11	PASS (both datasets)
122	YCB-V with synthetic renders	val PoE+orbit 12.74 vs 74.06 (real only); test PoE greedy 10.90, PoE+orbit 12.16	PASS
124	Official score, GT-crop reference	S1 (distilled + orbit) score LB/SL 0.227 / 0.224; S2 (joint PoE) 0.241 / 0.238; gate reproduces 11.21 / 11.09° and 0.185 / 0.183 m	descriptive; gate passed
127	Quantisation floor (no learning)	synth nearest-anchor mean / max: Tesseract L5 2.507 / 5.776; balanced Hopf 2.291 / 3.840; SPACE-HOP Hopf 6.057 / 14.973	trainings pending

13% / 28% / 23% for PoE. Clean synth also improves (DINOv3 4.57 vs 4.84, PoE 4.56 vs 4.85). Student + orbit is the best SPEED+ system. By DEC-023 it is reported as an enhancement, and the locked recipe stays EXP-109.

Zero-shot. The frozen SPEED+ trees do not transfer to other objects. On T-LESS the pre-set rule was literally met (PoE greedy pair error 95.6° vs 104.5° chance), but a post-hoc constant prediction does better (88.7°), because T-LESS objects are seen from above a table. On YCB-V the constant gives 57.8° against 95.9° and 101.6° for the trees. On SYMSOL-T, PoE + orbit beats chance (47.8° vs 53.7°) only because of the cube.

Other findings. SYMSOL-T orbit errors can fall below the leaf quantisation floor (2.31° mean, post hoc), because the tree may pick the leaf closest to any equivalent rotation. With canonical labels (A1), the weighted vote is far better than the orbit product (SYMSOL-T val 3.04 vs 9.73); the canonical label is not related across in-plane rotations by the roll transform, so the orbit views disagree.

Cost. At batch 1, PoE + orbit takes 278.5 ms per image, and most of it is MoGe-2 (84.2 ms) and its CPU consensus mask and post-processing (80.8 ms). DINOv3 single-view takes 14.7 ms (68.1 FPS). The tree head is faster than a flat head only at large hypothesis counts: at 131k hypotheses it took 1.2 vs 3.3 ms on an A400 and 1.7 vs 12.2 ms on CPU, but at SPACE-HOP's 3,072 anchors the flat head is faster, and on a large shared GPU at batch 1 the flat 131k-way head was about $3\times$ faster (0.40 vs 1.23 ms).

Gap to fine-tuned methods. On SYMSOL-T our best orbit result (DINOv3, A2 + orbit, 2.07° test, averaged over the five shapes) is about $4\times$ LiePose's [13] 0.55°, and on T-LESS R@2 is 5.7% against 34.9%. The depth-5 quantisation floor, the frozen backbone and LiePose's ground-truth visible masks are confounded in this gap. For translation, SPACE-HOP reports E_T 0.096 / 0.302 / 0.282 m with its own box source, which is not matched to our ground-truth crops.

Pending. Detected crops (EXP-124, completed). With the synthetic-only YOLO11s detector (mean IoU 0.986 /

0.764 / 0.884 on synthetic / lightbox / sunlamp; 54 lightbox misses fall back to the full image), the primary system reaches $3.88°/23.88°/16.42°$, $0.067/0.662/0.270$ m and a score of $0.081/0.523/0.333$, against $0.075/0.227/0.224$ with ground-truth crops (detection gap $+0.296$ lightbox, $+0.109$ sunlamp). The box-relative translation target passes box errors directly into E_T.

The following are pre-registered, but have no results yet: **EXP-119** (orbit distillation on T-LESS and SYMSOL-T), **EXP-126** (translation on T-LESS, SYMSOL-T and YCB-V against box-only) and the **EXP-127 trainings** (tree vs flat over the identical 131,072 leaves, and vs the balanced Hopf grid). EXP-128 (deployment and quantisation of the head) is also pending (Appendix J.4). If the same-grid flat head ties the tree, the hierarchy claim is withdrawn and the EXP-113 gain is attributed to the grid.

J.4. Deployment of the downstream head

Every application shares the same frozen backbones and the same inference algorithm; what is trained and shipped per application is only the downstream head: one tree per branch (DINOv3 and, for the PoE variant, MoGe-2) and, for 6D, the translation block. The head architecture is identical for every object and dataset, so its storage, latency and memory are properties of the design and are measured once. Its accuracy under reduced precision depends on the trained weights and is measured per application with each application's own evaluator and metric. This experiment (EXP-128) was pre-registered before any run; the frozen backbone is not quantised and its cost is reported separately (Appendix A).

Precisions. fp32 is the reference. bf16 (autocast) is the precision of every other result in this paper. fp16 uses autocast. int8 is weight-only: every matmul weight of the head (all linear layers including the attention projections, 20 tensors per tree, 99.3% of its parameters) is rounded to symmetric per-output-channel int8; activations, LayerNorms, embeddings and biases stay in fp32. Its accuracy is measured by simulating this rounding; its CPU speed is measured with PyTorch's dynamic int8 kernels, which convert the linear layers but not the attention in-projections. At fp32 the DINOv3 tree occupies about 9.1 MB (2,391,564 parameters); with int8 weights the ideal size is about a quarter of that.

Criterion. A precision's loss is *negligible* if its mean error differs from fp32 by no more than the larger seed standard deviation, per application, inference mode and split (the pass rule used throughout the paper).

J.5. Project history and decisions

Phase 0: geometry (EXP-000–003). EXP-000 fixed the SPEED+ pose convention $p_{\text{cam}} = R(q)\,p + r$ (keypoint boxes match with IoU 0.97 for R against 0.52 for $R^\top$). EXP-001 verified the Tesseract encoder on every label, and EXP-002/003 measured its precision and compared it with other grids. The gate set $L_{\max} = 5$ and kept a Hopf grid as the mandatory matched baseline.

Phases 1–2: branches and fusion (EXP-010–029). On a 9,500-image subset, a hierarchical MLP probe on DINOv3 `grid4` reached a synth mean of 27.7°, with real medians about $4\times$ worse. VGGT, the plan's geometric branch, was weaker everywhere and about $17\times$ slower, and MoGe-2 replaced it. A training-free consensus mask fixed MoGe-2's normals on real images. An equal-weight PoE of the two branches beat the best single branch (synth 25.9° vs 27.7°), and no reliability weighting or learned fusion beat it.

Phase 3: the tree at full data (EXP-030–036). Training on all 47,966 images cut the error about $4\times$ (MLP tree, PoE 6.24°). Of five decoder and target variants, the cross-attentive Transformer with geodesic soft targets (EXP-036) was best (PoE 5.540°; LB 26.9°, SL 36.6°) and was frozen. DINOv3 alone was then better on synth (5.35°) but far worse on real images. This exposed the selection-policy conflict that DEC-005 and DEC-006 settled in favour of the strict split policy.

Phase 4: continuous refinement and the flip audit (EXP-040–050, 104, 105). Residuals and Riemannian flows [2] improved synth precision slightly, flow-induced Tesseract masses did not help, and none of them fixed the real-domain errors. The flip audit (EXP-104) showed that these errors are 180° body-axis flips caused by visual near-symmetry: the right mode is among the tree's hypotheses but is not selected. EXP-105 confirmed that no scorer on the clean features selects it on real images. An adaptive-depth oracle (EXP-050) showed small synth headroom (0.51°), so adaptive depth became a low priority and was not pursued.

Real-domain phase (EXP-108–118). Four levers were proposed against mode selection. Augmentation plus orbit voting (EXP-109) cut the PoE errors from 26.91° / 36.66° to 12.31° / 12.28° (LB / SL), and the recipe was locked on it (DEC-018). Strengthening studies followed: tree vs flat (EXP-113), scoring rules (EXP-114), an augmentation ablation (EXP-115), two training-time alternatives to orbit voting (EXP-116, 117) and 6-DoF (EXP-118).

Cross-dataset phase (EXP-064, 106, 119, 122, 124, 126, 127). Per-object trees with the unchanged recipe worked on T-LESS and SYMSOL-T, failed on real-only YCB-V, and did not transfer zero-shot (EXP-064). Three fixes followed:

Table 28. **Negative and abandoned directions** (synth / LB / SL in degrees unless stated).

Direction	What was tried	Outcome
Geometric branch	VGGT tokens and maps (EXP-011, 012)	Synth mean 48.7 vs 27.7 for DINOv3; about 17× slower; replaced by MoGe-2.
Spectral fusion	four CASS-inspired options (EXP-017–020)	Worse masks or pose, e.g. spectral parts synth 53.0 vs 28.5; all rejected.
Masked DINOv3	mask-weighted DINOv3 pooling (EXP-016)	Synth 27.7 to 31.1; real changed by under a degree.
Fusion weighting	reliability weights (EXP-026); weighted PoE (EXP-110)	None beat equal PoE; $w^* = 0.8$ on val-aug made LB worse (32.0 vs 26.9).
Learned fusion	token concatenation (EXP-029)	Synth 5.341 equals DINOv3 alone; LB 40.8 vs 27.9.
Decoder variants	GRU decoder (EXP-031); beam search (EXP-034)	GRU missed the rule by 0.005; beam 8 changed synth by +0.006.
Tangent residual	at GT leaf (EXP-035) and predicted leaf (EXP-040)	Gains 0.055 and 0.13 on synth only; real moved by at most 0.03.
Continuous flows	cell-conditioned refiner (EXP-041); $SO(3)$ posterior (EXP-042, 111)	Synth better; real not fixed (LB 29.85 vs 26.88); posterior over-covers (0.98 at 0.9).
Flow masses	distil flow masses into the tree (EXP-043, 044)	Fused synth not improved (5.589 at β=0 vs 5.622); stopped by rule.
Path revision	discrete flow matching [6] (EXP-108)	7.22 vs 5.54; worse on both real domains.
Adaptive depth	oracle stopping depth (EXP-050)	Synth headroom 0.51; real gain only by hedging; controllers not run.
Flip selection	candidate expansion, templates, best fit (EXP-105)	Right flip present (LB oracle 9.8) but no scorer selects it.
Kurz corner tree	8 facets, corner targets (EXP-105 arm K)	Synth 5.20 vs 5.54, but LB 28.8, SL 39.1; excluded.
Finer tokens	16 × 16 DINOv3 grid (EXP-109 B)	Val-aug 30.51 vs 30.84, within seed std.
Rotation aug.	credited for flip repair after EXP-109	Refuted by EXP-115: photometric-only matches real (LB 12.29 vs 12.30).
Consistency loss	hierarchical roll transport (EXP-116)	Not better than a leaf-only equivariance loss (9.57 vs 9.43).
Canonical labels	A1 on symmetric objects (EXP-106)	Breaks orbit voting: SYMSOL-T orbit 9.73 vs 2.65 for A0.
Real-only YCB-V	per-object trees on real videos (EXP-064)	Test 69.4 (DINOv3+orbit); memorises videos; fixed by EXP-122.

Head module	Params	fp32 (MB)	fp16 (MB)	int8 (MB)
Tree, DINOv3 branch	2,391,564			
Tree, MoGe-2 branch	2,128,404			
Translation block	331,011			

Table 29. **Size of the downstream head** per module and precision (bytes of the saved weights). Blank cells: run not yet completed.

		GPU (ms/img)				CPU, 1 thread (ms/img)	
Head	Inference	fp32	bf16	fp16	peak MB	fp32	int8
DINOv3	1 view						
	orbit (4)						
PoE	1 view						
	orbit (4)						

Table 30. **Head-only latency** (ms per image, batch 1) and peak transient GPU memory, excluding the shared backbone. Orbit: 4 views × beam 8 plus scoring of 32 candidates. Shared GPU; times are upper bounds. Blank cells: run not yet completed.

Application	Inference	fp32	bf16	fp16	int8 (w)
T-LESS (A2, PoE)	1 view				
	orbit (4)				
SYMSOL-T (A2, DINOv3)	1 view				
	orbit (4)				
YCB-V (syn., DINOv3)	1 view				
	orbit (4)				
SPEED+ (distilled, PoE)	1 view				
	orbit (4)				

Table 31. **Accuracy of the head under reduced precision**: mean rotation error (°, symmetry-aware where applicable, real test sets; SPEED+ lightbox) of the validation-selected head per application, 3 seeds. Blank cells: run not yet completed.

symmetry-aware targets (EXP-106), synthetic YCB-V renders (EXP-122) and cross-dataset distillation (EXP-119). Matched comparisons with published work (EXP-124, 126) and the same-grid flat baseline (EXP-127) were added last. Along the way the subset-v1 feature caches of Phases 1–2 were deleted to free disk space (DEC-020), so those early runs cannot be re-evaluated without re-extracting features. Two incidents are worth knowing about. Nine SYMSOL-T MoGe-2 trees trained to NaN, because always-empty border cells had an sd of about 10^{-6} and overflowed fp16; they were retrained with an sd floor. A memory-guard false abort, caused by counting shared DataLoader pages per worker, stalled the EXP-124 queue for about 14.7 h.

Table 32 lists the decisions that shaped the final design.

J.6. Negative results

Table 28 lists the most important directions that failed or were abandoned. A common pattern runs through them: on SPEED+, almost every idea that read the clean-trained features improved synthetic precision but did not repair the real-domain flips. Only photometric augmentation combined with orbit voting did.

LoRA adaptation of the backbones (EXP-102) was cancelled by the user before any run, so all results use frozen

Table 32. **Key decisions** (from `outputs/DECISIONS.md`; all dates 2026).

DEC	Date	Decision
000	10-05	Train on synthetic only; select on synthetic val; real domains test-only.
001	10-08	MoGe-2 replaces VGGT as the geometric branch.
003	10-09	Fusion frozen as training-free equal-weight per-level PoE.
004 upd.	10-09	Transformer tree with geodesic soft targets (EXP-036) frozen.
006	10-10	Real-domain validation split (DEC-005) withdrawn; strict split policy restored.
008	10-13	Learned feature fusion not adopted; PoE kept.
012	10-13	Adaptive depth reduced to an oracle check; cross-dataset work after the recipe lock.
014	10-15	Augmentation, 16×16 tokens and orbit voting tested together as EXP-109.
018	10-16	Recipe lock: frozen features, one tree per branch, augmentation, PoE, orbit voting.
019	10-16	Strengthening studies: tree vs flat, scoring rules, augmentation ablation.
023	10-21	Recipe unchanged; orbit distillation reported as an enhancement.
024	10-27	Cross-dataset fixes: synthetic YCB-V, multi-path targets, cross-dataset distillation.
025	10-28	Core matched set: SPEED+ detector and official score, cross-dataset translation.
026	10-29	EXP-127: tree vs flat on the same 131,072 leaves.

backbones. The flow posterior is kept only for its density: its ground-truth log-likelihood on real images is much higher than the tree's (3.85 vs 0.98 nats on LB).

J.7. Claim boundaries

Claimed.

- A lightweight hierarchical rotation predictor on frozen features: one Tesseract-path Transformer per backbone, fused per level by a training-free PoE, with about 4.5 M trainable parameters. It is far more accurate than a count-matched flat Hopf classifier (EXP-113).
- Real-domain error on SPEED+ is a mode-selection problem: the large errors are 180° flips whose correct mode is already among the hypotheses (EXP-104, 105).
- Photometric augmentation plus orbit voting resolves most flips. EXP-109 fixes 62%/80% of the $> 150^\circ$ failures (LB/SL) and breaks 0.4% of good predictions; the claim is orbit consistency, not a specific scoring rule.
- Orbit distillation improves both the single-view tree and the orbit system (EXP-117), as an enhancement.
- 6-DoF with the SPACE-HOP translation formulation, trained jointly without hurting rotation (EXP-118).
- With per-object retraining, the recipe transfers to T-LESS and SYMSOL-T, and multi-path targets (A2) improve both while keeping orbit voting effective (EXP-106).

Not claimed.

- No "first" claims: hierarchical $SO(3)$ grids, the recursive tesseract [15], flow matching for pose, test-time voting and PoE all pre-exist.
- Not a better discretisation: per hypothesis the Tesseract is less precise than Hopf and Super-Fibonacci grids. Its head is faster only at large hypothesis counts (see Cost above).
- Hierarchy vs grid is open: the EXP-113 flat grid has a floor of 6.06° against 2.51°, and the gain is not attributed to hierarchy until EXP-127 runs.
- No claim that the orbit product beats weighted voting (EXP-114), and no hierarchical-transport claim (EXP-116).
- The flow models are not calibrated, and the path flow has no geometry claim.
- Not training-free and not zero-shot: the trees are trained, and outside SPEED+ they are retrained per object.
- MoGe-2 fusion helps on SPEED+ and T-LESS only.
- External comparisons are unmatched: SPEED+ uses ground-truth crops (the box alone gives 0.56 m translation error), with a synthetic-only detector the score rises to $0.523/0.333$ (detector IoU 0.76 on lightbox), and our T-LESS crops are not masked, unlike LiePose's [13].
- Experiments never evaluated together are not combined (*e.g.* the recipe with $w = 0.8$).